Irregular Repetition Slotted ALOHA with Multi-Antenna Reception: BP and MAP Thresholds and Converse Bound

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***Abstract*—Irregular repetition slotted ALOHA (IR-SA) is a promising random access scheme that exploits packet repetitions and iterative successive interference cancellation (SIC) to achieve high throughput. However, most existing studies focus on single-antenna reception, and the theoretical analysis of multi-antenna IR-SA over fading channels remains limited. This paper investigates IR-SA with multi-antenna reception over Rayleigh block-fading channels using iterative selection-combining-based SIC (SC-SIC) decoding. By applying the inclusion-exclusion principle, we derive exact closed-form expressions for the average decoding error probability $\bar{\epsilon}_{m,L}$ for $m = 1$ and $m = 2$ and develop a closed-form approximation for $m \geq 3$, both applicable to any finite number of receive antennas L. Based on these results, we formulate a density-evolution analysis and characterize the belief-propagation (BP) thresholds of multi-antenna IR-SA. The analysis further reveals a performance tradeoff between SC-SIC and maximal-ratio-combining-based SIC. We also characterize the maximum a posteriori (MAP) thresholds and derive a converse bound, and show that spatial coupling drives the BP threshold toward the MAP threshold and the corresponding converse bound, demonstrating threshold saturation.**

***Index Terms*—Belief propagation threshold, irregular repetition slotted ALOHA, multi-antenna reception, random access, Rayleigh fading, successive interference cancellation.**

I. Introduction

Massive machine-type communication (mMTC) is expected to be a key enabler of sixth-generation (6G) communication systems [1]. In mMTC systems, a large number of devices are typically connected to each access point (AP), while only a small fraction of them become active sporadically and unpredictably [2]. Therefore, random access (RA) schemes [3] are more suitable than scheduled access for uplink transmissions, as they can reduce signaling overhead. Among various RA schemes to efficiently support mMTC, those evolving from slotted ALOHA (SA) [4] have garnered significant attention, as they allow devices to transmit in an 'arrive-and-go' manner.

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In conventional SA, each active device among the N_{T} devices transmits its packet in a slot to an AP, where collided packets are simply discarded according to the collision channel model. Consequently, the packet loss rate (PLR) of SA grows rapidly with the offered traffic per slot. To mitigate these collisions, contention resolution diversity slotted ALOHA (CRDSA) [5] and irregular repetition slotted ALOHA (IR-SA) [6]–[13] employ packet repetition and inter-slot successive interference cancellation (SIC) to iteratively remove decoded packets from collided slots.

In IR-SA systems, each of the N_{T} devices becomes active independently with an activation probability $\pi \ll 1$. Each active device selects a repetition degree according to a degree distribution $\Lambda(x)$ and transmits the corresponding packet replicas over randomly selected slots within a frame of M slots. For asymptotic analysis ($N_{\mathrm{T}}, M \to \infty$), density evolution (DE) has been employed to determine the maximum average normalized offered traffic G that maintains the PLR below a target threshold. In particular, IR-SA over Rayleigh block-fading channels with single-antenna reception ($L = 1$) has been analyzed under threshold-based decoding with intra-slot SIC [6]. Assuming that a packet is decoded when its signal-to-interference-plus-noise ratio (SINR) exceeds a prescribed decoding threshold η_0, corresponding to a spectral efficiency $R \geq 1$, an exact closed-form expression for the average decoding error probability $\bar{\epsilon}_{m,1}$ in an m-packet collision was derived for any finite m [6]. This probability enables a rigorous DE-based analysis of IR-SA systems and the derivation of the belief propagation (BP) threshold [6].

The aforementioned IR-SA schemes [6]–[13] assume a single-antenna AP. In contrast, several studies have introduced spatial-diversity reception into IR-SA and related graph-based RA schemes using multiple antennas or receivers. However, when explicit fading distributions, such as Rayleigh fading, are taken into account, analytical characterization requires the joint treatment of the SINR and SIC-dependent decoding events across the receive branches. The complexity of this characterization grows significantly with both the collision size and the number of receive branches. Consequently, existing analyses of IR-SA and related graph-based RA schemes with spatial-diversity reception primarily rely on simplified packet-erasure models [14]–[19] or numerical simulations [20].

For instance, spatial-diversity reception has been introduced into IR-SA to enable joint *spatio-temporal* SIC [14]. Under a geographic-proximity collision model, a packet involved in a

collision at one receiver may be decoded as a singleton at another receiver and subsequently canceled from the collided observations from the other receivers. Its replicas are also removed from the other slots, thereby enabling further decoding across both receivers and slots. Related coverage-based collision models have also been used to analyze multiple-base-station frameless slotted ALOHA, which retains the packet-repetition and graph-based SIC structure of IR-SA [17]–[19]. A more general probabilistic abstraction for coded SA with multiple receivers was developed based on the Poisson-receiver framework [15], [21]. In this framework, the underlying physical-layer receiver is represented by a success-probability function under a Poisson offered load, enabling a tractable analysis of *spatio-temporal* SIC. Nevertheless, these abstraction-based approaches do not explicitly model random fading coefficients or the resulting SINRs and therefore do not derive the average decoding error probability $\bar{\epsilon}_{m,L}$ as a function of the collision size m and the number of receive branches L.

More recently, building on multi-AP cooperative SA [14], Qin et al. proposed non-orthogonal multiple access (NOMA)-based cooperative IR-SA that combines transmit-power diversity with intra-slot, inter-slot, and inter-AP SIC [22]. Their channel model explicitly incorporates distance-dependent path loss, lognormal shadowing, and small-scale multipath fading. Although the asymptotic PLR is analyzed via DE, the probability that a packet is successfully decoded from a collision within a slot, which is analogous to $1-\bar{\epsilon}_{m,L}$ in our framework, is evaluated numerically using Monte Carlo simulations because the decoding events across APs are correlated and the iterative cooperative SIC results in nonlinear, dynamically evolving SINRs. Thus, even with an explicit physical-layer channel model, analytically characterizing the decoding reliability within a slot remains challenging.

More directly related to our setting, Srivatsa *et al.* investigated IR-SA over Rayleigh block-fading channels with a multi-antenna AP [23]. Their model explicitly characterizes the received signal vectors and the resulting SINRs, and employs maximal-ratio combining (MRC) to coherently combine the received signal of each packet across the antenna branches. MRC is incorporated into the intra-slot SIC process, while decoded replicas are removed from the other slots through inter-slot SIC. In [23, Theorem 3], under perfect channel state information (CSI), the average successful decoding probability θ_m of a reference packet starting from an m-packet collision, corresponding to $1-\bar{\epsilon}_{m,L}$ in our notation, was characterized under MRC-based SIC (MRC-SIC). An exact expression was derived for a singleton slot ($m=1$), whereas closed-form expressions and approximations for collided slots ($m\geq 2$) were developed based on the large-antenna regime. In particular, for $m\geq 3$, the normal approximation replaces the maximum SINR in the first SIC iteration with the SINR of a representative packet and approximates the aggregate multiuser interference experienced by that packet as Gaussian. Moreover, the joint decoding events associated with the second and subsequent SIC iterations are not explicitly evaluated. Therefore, an exact characterization of the average decoding error probability $\bar{\epsilon}_{m,L}$ for arbitrary finite m and L remains an open problem.

In the asymptotic analysis of IR-SA, the maximum a posteriori (MAP) threshold and the converse bound serve as theoretical limits for the BP threshold, similar to the decoding analysis of low-density parity-check (LDPC) codes over a binary erasure channel (BEC) [24]. Packet-wise MAP decoding, analogous to its bit-wise counterpart in LDPC codes, maximizes the individual packet posterior probabilities to establish the MAP threshold, G^{MAP}. This threshold has been derived over collision [8] and Rayleigh fading channels [7]. Representing the highest achievable performance under unconstrained decoding complexity, it serves as a fundamental benchmark for practical schemes such as BP. The converse bound G^{C}, in contrast, provides an information-theoretic upper bound on the achievable load, irrespective of the decoding algorithm. Its practical significance therefore lies in quantifying how closely the MAP or BP threshold approaches this fundamental limit; a small gap indicates that the corresponding bound is tight. Converse bounds have been investigated over the collision channels [8], [11], [25], Gaussian channels [13], and Rayleigh fading channels [7], [12]. Building on the DE characterization, the converse bound can be derived using an extrinsic information transfer (EXIT) chart area argument, while the MAP threshold can be characterized through trial-entropy analysis [8], [25], [26]. To the best of our knowledge, however, neither the MAP threshold nor the converse bound has been characterized for IR-SA systems under multi-antenna reception.

In this paper, we derive analytical expressions for the average decoding error probability $\bar{\epsilon}_{m,L}$ of IR-SA systems with L-antenna reception over Rayleigh fading channels. Our analysis considers selection-combining-based SIC (SC-SIC), in which the receiver selects the strongest signal among the L antennas. Our main contributions are as follows:

1) **Exact Closed-Form Analysis:** We apply the inclusion-exclusion principle to multi-antenna reception with intra- and inter-antenna SIC to derive exact closed-form expressions for the average decoding error probability $\bar{\epsilon}_{m,L}$ for collision sizes $m=1$ and $m=2$ and any finite L. Although the probability derivations initially require separate treatments of $R\geq 1$ and $0<R<1$, we show that the final expressions can be unified for $R>0$, thereby simplifying the probability analysis. Moreover, the regime $0<R<1$ is particularly relevant to SA, since multi-device interference typically necessitates low-rate channel coding together with low-order modulation, resulting in spectral efficiencies below one in practical physical-layer scenarios [27].
2) **Closed-Form Approximation for Higher-Order Collisions:** For collision sizes $m\geq 3$, we develop a closed-form approximation for $\bar{\epsilon}_{m,L}$ by combining the exact first-step decoding probability for an arbitrary collision size over the entire range $R>0$ with the exact two-packet result. After one packet is decoded and removed by inter-antenna SIC, the residual collision is approximated as a new $(m-1)$-packet collision. This construction captures the decoding of the reference packet at later SIC stages while yielding a closed-form

expression for the decoding error probability for any collision size $m \geq 3$, any finite L, and $R > 0$.

3) **Density Evolution and Performance Trade-offs:** Based on the average decoding error probability $\bar{\epsilon}_{m,L}$, we establish a slot-node (SN)-level abstraction that encapsulates all multi-antenna processing within each slot node. Specifically, the effect of L-antenna reception is fully captured by the SN decoding success probability $1 - \bar{\epsilon}_{m,L}$, while the graph-level DE recursion retains the same structural form as in the single-antenna case. This enables the BP threshold of IR-SA with multi-antenna reception to be characterized for any finite L. Furthermore, the resulting BP thresholds elucidate the performance trade-offs between SC-SIC and MRC-SIC [23].
4) **Bounds and Threshold Saturation:** Building on the EXIT-area and trial-entropy frameworks developed in our previous single-antenna study [7], we extend the converse-bound and MAP-threshold analyses from $L = 1$ to any finite number of receive antennas. The effect of multi-antenna reception is incorporated through the L-antenna SN decoding function $\chi_L(\cdot)$, which is determined by the decoding probability $1 - \bar{\epsilon}_{m,L}$ ($m \geq 1$). Specifically, the converse bound is derived using the EXIT-area argument, whereas the MAP threshold is characterized through trial-entropy analysis. The resulting MAP thresholds closely approach the corresponding converse bounds, while spatial coupling enables the BP thresholds to approach the MAP thresholds, demonstrating threshold saturation under multi-antenna reception.

II. System model

A. IR-SA Access Protocol with Multi-Antenna Reception

We consider an IR-SA system with multi-antenna reception, where N_{T} single-antenna devices communicate with an AP equipped with L antennas over a frame of M time slots. Let $\mathcal{L} = \{1, 2, \ldots, L\}$ denote the set of receive antenna indices. The antennas at the AP are spaced sufficiently far apart to ensure independent fading values. In each frame, each device becomes active independently with probability $\pi \ll 1$ and, if active, generates one message that is encoded and modulated into a packet at a spectral efficiency R using a common channel code $\mathcal{C}$ and a modulation scheme. Each active device selects a repetition degree l with probability Λ_l according to the distribution $\{\Lambda_l\}_{l=1}^{l_{\max}}$. It transmits l replicas of its packet in l randomly selected slots within a frame.

We define the normalized population size as $\alpha \triangleq N_{\mathrm{T}}/M$, which is held constant in the asymptotic regime. Since $\mathbb{E}[N_{\mathrm{a}}] = \pi N_{\mathrm{T}}$, the normalized offered traffic is defined as $G \triangleq \frac{\mathbb{E}[N_{\mathrm{a}}]}{M} = \pi\alpha$, which represents the average number of packets transmitted per slot when PLR is lower than a target value. We are interested in the asymptotic ($N_{\mathrm{T}}, M \to \infty$) traffic G, which depends on probability distribution $\{\Lambda_l\}_{l=1}^{l_{\max}}$, given the average received signal-to-noise ratio (SNR), the spectral efficiency R and the number of antennas L.

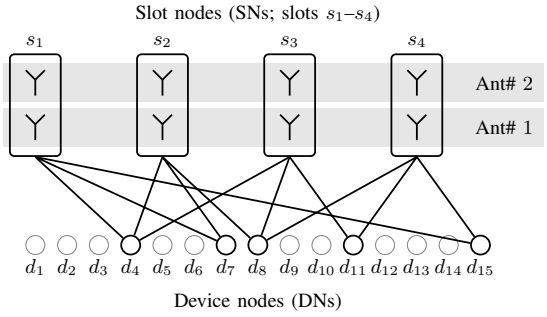


Fig. 1. Bipartite graph representation of multi-antenna IR-SA with $L = 2$, $M = 4$, $N_{\mathrm{T}} = 15$, $N_{\mathrm{a}} = 5$, and $\pi = N_{\mathrm{a}}/N_{\mathrm{T}} = 1/3$.

B. Bipartite Graph and Degree Distributions

As illustrated in Fig. 1, the multi-antenna IR-SA protocol outlined in Sec. II-A can be characterized in a manner analogous to conventional single-antenna IR-SA [9], [10]. Specifically, it is modeled as a bipartite graph $\mathcal{G} = (\mathcal{D}, \mathcal{S}, \mathcal{E})$. Here, $\mathcal{D}$ denotes the set of N_{T} device nodes (DNs), $\mathcal{S}$ denotes the set of M slot nodes (SNs), and $\mathcal{E}$ represents the set of edges. An edge connects a DN $d_i \in \mathcal{D}$ to an SN $s_j \in \mathcal{S}$ if and only if the i-th device selects the j-th slot for transmission.

Importantly, even with multiple receive antennas, all multi-antenna processing, including intra- and inter-antenna SIC, is performed locally within each slot node. Therefore, the bipartite graph is determined solely by the device transmission patterns and retains the same structure as in the single-antenna case, consistent with prior work [23]. This observation allows the entire multi-antenna receiver operation within an m-degree slot node to be abstracted by its average decoding error probability $\bar{\epsilon}_{m,L}$. Consequently, the graph-level DE recursion retains the same structural form as for single-antenna IR-SA, with the effect of L-antenna reception fully encapsulated in the slot-node decoding success probability $(1 - \bar{\epsilon}_{m,L})$ (see (30)).

The number of edges connected to a DN or SN is defined as the node degree. In this framework, a DN has degree l (i.e., it has l edges connected to SNs) with probability Λ_l. The set $\{\Lambda_l\}_{l=1}^{l_{\max}}$ is referred to as the DN degree distribution, where the average number of replicas per active device is given by

$$\bar{d} \triangleq \Lambda'(1) = \textstyle\sum_{l=1}^{l_{\max}} l\Lambda_l. \tag{1}$$

As $N_{\mathrm{T}} \to \infty$, the SN degree distribution is Poisson with $\Psi_n = \exp(-\alpha\bar{d})(\alpha\bar{d})^n/n!$ [9], [10].

The degree distributions are also defined from an edge-perspective in polynomials [10]: $\lambda(x) = \sum_{l=1}^{l_{\max}} \lambda_l x^{l-1}$ with $\lambda_l = \frac{l\Lambda_l}{\sum_{l=1}^{l_{\max}} l\Lambda_l}$ and

$$\rho(x) = \textstyle\sum_{n=1}^{N_{\mathrm{T}}} \rho_n x^{n-1} = \exp(-\alpha\bar{d}(1-x)), \tag{2}$$

with Poisson distribution

$$\rho_n = \frac{n\Psi_n}{\sum_{n=1}^{N_{\mathrm{T}}} n\Psi_n} = \exp(-\alpha\bar{d})\frac{(\alpha\bar{d})^{n-1}}{(n-1)!},\ n = 1, 2, \ldots. \tag{3}$$

C. Spatio-Temporal SIC Decoding in Rayleigh Block Fading

Decoding at the AP is an iterative decoding process, integrating both intra- and inter-slot as well as intra- and inter-antenna SICs.

1) Intra-Slot SIC: Assuming m packet replica collisions, the resultant L superimposed signals (one per antenna) within the slot are given by

$$\boldsymbol{y}_i = \sum_{k\in\mathcal{K}} h_{k,i}\boldsymbol{x}_k + \boldsymbol{z}_i, \quad i = 1, 2, \ldots, L, \tag{4}$$

where $\mathcal{K}$ ($|\mathcal{K}| = m$) is the set of active devices. Here, device k is active and has accessed the slot. The vector $\boldsymbol{x}_k \in \mathcal{C}$ represents its codeword (or packet) of code $\mathcal{C}$. The vector $\boldsymbol{z}_i \sim \mathcal{CN}(\boldsymbol{0}, \sigma^2 I)$ is an additive circularly symmetric complex Gaussian (CSCG) noise with mean $\boldsymbol{0}$ and diagonal covariance matrix $\sigma^2 I$ with noise power σ^2 and identity matrix I. The channel coefficient $h_{k,i}$ expresses the fading of the channel from the active device k to Ant# i of AP during the slot, which remains constant within a slot and varies independently across slots. For Rayleigh block fading channels, the received SNR $\gamma_{k,i} \triangleq P|h_{k,i}|^2/\sigma^2$ is exponentially distributed with $p_\gamma(x) = 1/\bar{\gamma} \cdot e^{-\frac{x}{\bar{\gamma}}}$, $x > 0$, where $\bar{\gamma}$ is the average received SNR.

Decoding at the AP within a slot involves an iterative process alternating between intra-antenna and inter-antenna SICs. Similar to [6], [7], [12], we assume that the AP has perfect knowledge of the CSI and the number of colliding packets. We consider SC-SIC, in which SIC is performed independently at each antenna, and the antenna branch with the strongest received signal is selected for packet decoding.

a) Intra-Antenna SIC: For an m-packet collision at Ant# i, assuming $\gamma_{1,i} > \gamma_{2,i} > \cdots > \gamma_{m,i}$, the SINR of the packet from device k is given by

$$\eta_{k,i} \triangleq \frac{P|h_{k,i}|^2}{\sigma^2 + \sum_{u=k+1}^{m} P|h_{u,i}|^2} = \frac{\gamma_{k,i}}{1 + \sum_{u=k+1}^{m} \gamma_{u,i}}. \tag{5}$$

Assuming threshold-based decoding, the packet $\boldsymbol{x}_k$ is successfully decoded if the SINR $\eta_{k,i}$ exceeds the threshold η_0, i.e.,

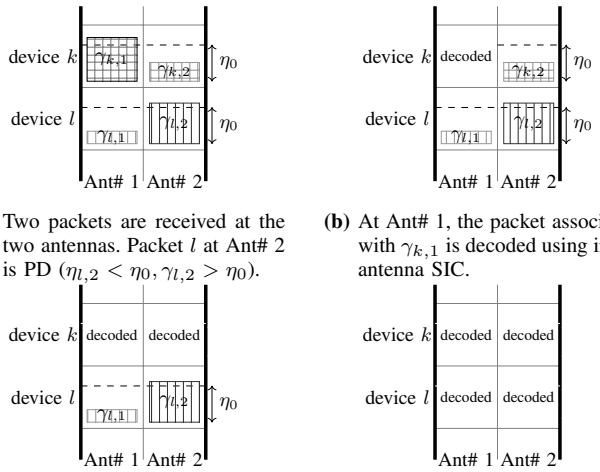
$$\Pr\{\text{packet } \boldsymbol{x}_k \text{ decoded}\} = \begin{cases} 1, & \eta_{k,i} > \eta_0, \\ 0, & \eta_{k,i} \le \eta_0, \end{cases} \tag{6}$$

where η_0 satisfies $R = \log_2(1 + \eta_0)$. Upon successful decoding, the packet is removed from the received signal $\boldsymbol{y}_i$. Subsequent decoding proceeds with the interference-removed signal. This iterative process is termed *intra-antenna* SIC.

An undecoded packet $\boldsymbol{x}_k$ with SNR above η_0, i.e., $\gamma_{k,i} > \eta_0$, is termed a *potentially decodable (PD)* packet [28], because it may become decodable after one or more interfering packets $\boldsymbol{x}_{j\neq k}$ are removed from signal $\boldsymbol{y}_i$.

b) Inter-Antenna SIC: The previously described intra-antenna SIC process enables the successful decoding of packets. These decoded packets are then canceled from the receive signals of other antennas within the same slot. This procedure is referred to as *inter-antenna* SIC.

The above two processes are repeated across all antenna elements until no further decoding is possible for any packet at any antenna element. An example of the intra- and inter-antenna SIC process is illustrated in Fig. 2. This iterative decoding procedure is referred to as *intra-slot* SIC due to its execution within a slot.



(a) Two packets are received at the two antennas. Packet l at Ant# 2 is PD ($\eta_{l,2} < \eta_0, \gamma_{l,2} > \eta_0$).

(b) At Ant# 1, the packet associated with $\gamma_{k,1}$ is decoded using intra-antenna SIC.

(c) Since $\gamma_{l,1} < \eta_0$, Ant# 1 does not perform decoding. The $\gamma_{k,2}$ associated with device k is canceled using inter-antenna SIC.

(d) Since $\gamma_{l,2} > \eta_0$, the packet of device l is decoded, and $\gamma_{l,1}$ associated with device l is canceled using inter-antenna SIC.

Fig. 2. An example of intra- and inter-antenna SIC decoding within a two-packet slot. The AP is equipped with $L = 2$ antennas.

2) Inter-Slot SIC: Following intra-slot SIC, once packets are successfully decoded, their replicas are removed from the specific slots identified by those packets. This processing, termed *inter-slot SIC*, facilitates subsequent intra-slot SIC on the remaining signals in those slots.

The intra- and inter-slot SIC processes can be modeled as the successive removal of edges in a bipartite graph $\mathcal{G}$. When a packet from a device (or a DN) is successfully decoded at least one antenna in a slot (or a SN), the corresponding edge connected to the SN is removed via intra-slot SIC. Subsequently, all l edges incident to the same DN are removed through inter-slot SIC. This process iterates until no further edges can be eliminated.

III. Probabilities in a Two-Packet Collision (Single-Antenna Case)

In this section, we derive closed-form expressions for the probabilities that two packets are received in distinct SNR regions. These expressions will be used for computation of the average decoding error probability $\bar{\epsilon}_{m,L}$ in Sec. IV.

A. Intra-Slot SIC

We consider a two-device fading channel with single-antenna reception, where the received signal at a slot is given by $\boldsymbol{y} = \sum_{k\in\mathcal{K}} h_k \boldsymbol{x}_k + \boldsymbol{z}$. Here, $\mathcal{K} = \{k, l\}$ denotes the set of active devices with $|\mathcal{K}| = 2$. The signal vector $\boldsymbol{x}_k$ and the noise vector $\boldsymbol{z}$ follow the definitions provided in Section II-C. The channel coefficient h_k expresses the fading of the channel from the active device k to the single antenna AP during the slot. The SNR of the signal received from the active device k at the single antenna AP is expressed as $\gamma_k = P|h_k|^2/\sigma^2$. The SNR γ_k follows an exponential distribution. We summarize the possible single-antenna reception events for the packets sent from device k and device l, as shown in Table I.

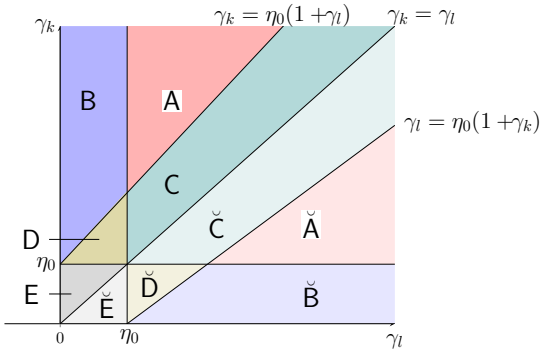


Fig. 3. The SIC feasible region in a two-packet collision ($m = 2$) between devices k and l under $R \geq 1$.

Before deriving the probability of two packets being received in distinct SNR regions, we define $\Gamma = \{(\gamma_k, \gamma_l)|\gamma_k \in \Gamma_k, \gamma_l \in \Gamma_l\}$ as a probability region, with $\Gamma_k, \Gamma_l \subseteq \mathcal{R}^+$ being subsets of the non-negative real numbers $\mathcal{R}^+$.

For the two independent RVs γ_k and γ_l, the probability of being in region Γ is given by

$$P_\Gamma = \Pr\{(\gamma_k, \gamma_l)|\gamma_k \in \Gamma_k, \gamma_l \in \Gamma_l\} = \int_{\Gamma_l} p(\gamma_l) \int_{\Gamma_k} p(\gamma_k) \mathrm{d}\gamma_k \mathrm{d}\gamma_l.$$

The probability expressions are divided into two cases: $R \geq 1$ and $R < 1$, as detailed below.

B. Closed-Form Expressions for the Probabilities ($R \geq 1$)

In the case of $R \geq 1$, Fig. 3 illustrates the SNR regions A to E ($\gamma_k > \gamma_l$) along with their symmetric counterparts $\breve{\mathsf{A}}$ to $\breve{\mathsf{E}}$ ($\gamma_k < \gamma_l$). The events listed in Table I are defined.

For $\gamma_k > \gamma_l$, the probabilities for regions A to E are:

$$P_\mathsf{A} = \Pr\{\mathsf{A} : (\gamma_k, \gamma_l)|\gamma_k > \eta_0(1+\gamma_l), \gamma_l > \eta_0\} \tag{7a}$$
$$P_\mathsf{B} = \Pr\{\mathsf{B} : (\gamma_k, \gamma_l)|\gamma_k > \eta_0(1+\gamma_l), \gamma_l \leq \eta_0\} \tag{7b}$$
$$P_\mathsf{C} = \Pr\{\mathsf{C} : (\gamma_k, \gamma_l)|\gamma_l < \gamma_k \leq \eta_0(1+\gamma_l), \gamma_l > \eta_0\} \tag{7c}$$
$$P_\mathsf{D} = \Pr\{\mathsf{D} : (\gamma_k, \gamma_l)|\eta_0 < \gamma_k \leq \eta_0(1+\gamma_l), \gamma_l \leq \eta_0\} \tag{7d}$$
$$P_\mathsf{E} = \Pr\{\mathsf{E} : (\gamma_k, \gamma_l)|\gamma_l < \gamma_k \leq \eta_0, \gamma_l \leq \eta_0\}. \tag{7e}$$

For $\gamma_k < \gamma_l$, the probabilities for regions $\breve{\mathsf{A}}$ to $\breve{\mathsf{E}}$ are defined analogously. In Appendix I-A, we derive the closed-form expressions for probabilities P_A through P_E while excluding $P_{\breve{\mathsf{A}}}$ and $P_{\breve{\mathsf{E}}}$ due to symmetry.

C. Closed-Form Expressions for the Probabilities ($R < 1$)

In the case of $R < 1$, the decoding threshold η_0 is less than 1, i.e., $\eta_0 < 1$. This implies that the two lines $\gamma_k = \gamma_l$ and $\gamma_k = \eta_0(1+\gamma_l)$ (or $\gamma_l = \eta_0(1+\gamma_k)$) intersect at a point denoted by $(\frac{\eta_0}{1-\eta_0}, \frac{\eta_0}{1-\eta_0})$ (see Fig. 4), where $\frac{\eta_0}{1-\eta_0}$ is the unique solution to equation $\gamma_k = \eta_0(1+\gamma_k)$ (or $\gamma_l = \eta_0(1+\gamma_l)$).

This implies that the probability that both packets are decodable over the entire γ_k–γ_l plane is given by

$$\Pr\left\{\mathsf{A}' \cup \breve{\mathsf{A}}' \cup \mathsf{F}' \cup \breve{\mathsf{F}}'\right\} = (P_{\mathsf{A}' \cup \mathsf{F}' \cup \breve{\mathsf{F}}'} - P_{\breve{\mathsf{F}}'}) + (P_{\breve{\mathsf{A}}' \cup \mathsf{F}' \cup \breve{\mathsf{F}}'} - P_{\mathsf{F}'}).$$

Note that the ranges of the double integral used to compute $P_{\mathsf{A}' \cup \mathsf{F}' \cup \breve{\mathsf{F}}'}$ are the same as those in (7a).

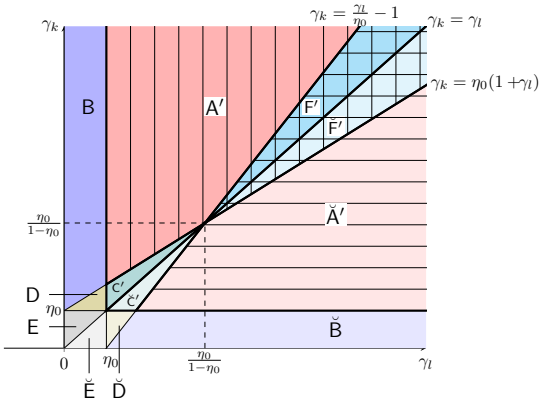


Fig. 4. The SIC feasible region in a two-packet collision between devices k and l in a slot under $0 < R < 1$. Vertical lines illustrate the region $(\mathsf{A}' \cup \mathsf{F}' \cup \breve{\mathsf{F}}')$, while horizontal lines represent the region $(\breve{\mathsf{A}}' \cup \mathsf{F}' \cup \breve{\mathsf{F}}')$.

To treat both ranges of R in a unified manner, as in the case of $R \geq 1$, we give the probabilities for the case $\gamma_k > \gamma_l$:

$$\begin{aligned} P_{(\mathsf{A}' \cup \mathsf{F}')} =& P_{\mathsf{A}' \cup \mathsf{F}' \cup \breve{\mathsf{F}}'} - P_{\breve{\mathsf{F}}'} \\ =& \Pr\{(\gamma_k, \gamma_l)|\gamma_k > \eta_0(1+\gamma_l), \gamma_l > \eta_0\} \\ & - \Pr\left\{(\gamma_k, \gamma_l)|\eta_0(1+\gamma_l) < \gamma_k < \gamma_l, \gamma_l > \frac{\eta_0}{1-\eta_0}\right\}, \end{aligned} \tag{8}$$

$$P_{\mathsf{C}'} = \Pr\left\{(\gamma_k, \gamma_l)|\gamma_l < \gamma_k < \eta_0(1+\gamma_l), \eta_0 < \gamma_l < \frac{\eta_0}{1-\eta_0}\right\}. \tag{9}$$

The probabilities for regions $(\breve{\mathsf{A}}' \cup \breve{\mathsf{F}}')$ and $\breve{\mathsf{C}}'$ ($\gamma_k < \gamma_l$) can be derived in a similar manner, and are omitted for brevity. We derive the closed-form expressions for $P_{(\mathsf{A}' \cup \mathsf{F}')}$ and $P_{\mathsf{C}'}$ in Appendix I-B. Note that the closed-form expressions for P_B, P_D, and P_E remain invariant with respect to R.

Since $P_{(\mathsf{A}' \cup \mathsf{F}')}$ and P_A represent the probability that two packets are successfully decoded, we adopt the unified notation P_A and extend it to the regime $0 < R < 1$. In this range, P_A follows the expression originally derived for $P_{(\mathsf{A}' \cup \mathsf{F}')}$, whereas for $R \geq 1$, it follows the expression previously denoted by P_A. Likewise, the probability P_C is extended to $0 < R < 1$ by employing its counterpart $P_{\mathsf{C}'}$ defined for this regime.

Before concluding this section, it is worth noting that although the above derivations are separated into two cases ($R \geq 1$ and $0 < R < 1$), a key outcome is that the final expressions can be unified for $R > 0$, with only some intermediate formulas retaining the two-case form. An advantage of this result is that it allows the derivation of average error probability $\bar{\epsilon}_{m,L}$ and, consequently, the BP thresholds, MAP thresholds, and converse bounds, under the general condition $R > 0$ without distinguishing between the two cases. While many IR-SA studies focus solely on the $R \geq 1$ regime (e.g., [6], [7], [12]), practical systems often operate with $0 < R < 1$ due to multi-device interference and the need for low-rate channel coding together with low-order modulation [27].

TABLE I
EVENTS AT A SINGLE ANTENNA, WHERE P_k AND P_l ARE PACKETS SENT BY DEVICES k AND l, RESPECTIVELY.

region	event
A ($\breve{\mathsf{A}}$)	P_k (P_l) is decoded first and then P_l (P_k) is decoded.
B ($\breve{\mathsf{B}}$)	P_k (P_l) is decoded, while P_l (P_k) is undecoded and not PD.
C, $\breve{\mathsf{C}}$	P_k and P_l are PD.
D ($\breve{\mathsf{D}}$)	P_k (P_l) is undecoded and PD; P_l (P_k) is undecoded and not PD.
E, $\breve{\mathsf{E}}$	P_k and P_l are undecoded and not PD.

IV. AVERAGE DECODING ERROR PROBABILITY IN A SLOT

Consider a slot t with m packets received by L antennas, as in (4), and select one packet as the reference. We derive exact closed-form expressions for $\bar{\epsilon}_{m,L}$ for $m=1,2$ under intra-slot SIC performed across and within antennas. Building on these exact results, we then formulate a closed-form approximation for $m \geq 3$. These results provide the analytical basis for the DE framework developed in the next section.

In threshold-based decoding, a single decoding step entails the recovery of a packet at any of the L antennas, followed by its interference cancellation across all antennas. Consequently, resolving an m-packet collision may require up to m successive steps. Let $D(m,r,L)$ denote the probability that the reference packet is successfully decoded exactly at the r-th step. The average decoding probability is then given by $D(m,L)=\sum_{r=1}^{m} D(m,r,L)$. Thus, the average decoding error probability for the reference packet, given $m-1$ interferers within a slot and L receive antennas, is expressed as

$$\bar{\epsilon}_{m,L} = 1 - D(m,L) = 1 - \sum_{r=1}^{m} D(m,r,L). \tag{10}$$

In deriving the results for the L-antenna case, we rely on the following fundamental facts:

(i) RVs $\gamma_{k,i}$ are independent and identically distributed.

(ii) The inclusion-exclusion principle for computing the probability of the union of L sets $A_1,\ldots,A_L$ [29]:

$$\Pr\left(\bigcup_{\ell=1}^{L} A_\ell\right) = \sum_{\emptyset\neq\mathcal{S}\subseteq\{1,\ldots,L\}} (-1)^{|\mathcal{S}|+1} \Pr\left(\bigcap_{\ell\in\mathcal{S}} A_\ell\right). \tag{11}$$

(iii) The binomial theorem: $\sum_{k=0}^{n}\binom{n}{k}x^{n-k}y^k=(x+y)^n$.

A. Exact Error Probability for $m \leq 2$

1) Closed-form Expression for a Single-Packet Slot ($m=1$): The probability that the packet of device k, which is the only packet received in the slot, is successfully decoded by at least one of the antennas, is given by

$$D(1,1,L) = 1 - \prod_{i=1}^{L}(1-\Pr\{\gamma_{k,i}>\eta_0\}) = 1-(1-e^{-\frac{\eta_0}{\bar{\gamma}}})^L. \tag{12}$$

2) Closed-form Expression for a Two-Packet Slot ($m=2$):

a) $D(2,1,L)$: We derive the probability that the reference packet is successfully decoded in the first threshold-based decoding step. This probability is expressed as

$$D(2,1,L) = \frac{1}{2}\Pr\{\Upsilon\} = \frac{1}{2}\Big(1-\{1-2(P_{\mathsf{A}}+P_{\mathsf{B}})\}^L\Big). \tag{13}$$

Here, $\Pr\{\Upsilon\}$ denotes the probability that either P_k or P_l is decodable at any of the L antennas in the first step. Note that the reference packet is either P_k or P_l with probability $1/2$. For simplicity, we first consider the case where the reference packet P_k (or P_l) is decoded at a specific antenna $i\in\mathcal{L}$ before P_l (or P_k). The probability of this event is given by $\Pr\{\Upsilon_i\} = \Pr\{\mathsf{A}_i\}+\Pr\{\mathsf{B}_i\}+\Pr\{\breve{\mathsf{A}}_i\}+\Pr\{\breve{\mathsf{B}}_i\} = 2(P_{\mathsf{A}}+P_{\mathsf{B}})$. Consequently, the probability that the reference packet is first decoded at any antenna in $\mathcal{L}$ is $\Pr\{\Upsilon\}=\Pr\{\bigcup_{i\in\mathcal{L}}\Upsilon_i\}$. With the inclusion-exclusion principle (11) and the binomial theorem, we obtain

$$\begin{aligned}
\Pr\{\Upsilon\} = \Pr\{\bigcup_{i\in\mathcal{L}}\Upsilon_i\} &= \sum_{\emptyset\neq\mathcal{S}\subseteq\mathcal{L}}(-1)^{|\mathcal{S}|+1}\Pr\left(\bigcap_{i\in\mathcal{S}}\Upsilon_i\right)\\
&= \sum_{\ell=1}^{L}\sum_{\substack{\mathcal{S}\subseteq\mathcal{L}\\|\mathcal{S}|=\ell}}(-1)^{\ell+1}\Pr\left(\bigcap_{i\in\mathcal{S}}\Upsilon_i\right)\\
&= \sum_{\ell=1}^{L}(-1)^{\ell+1}\binom{L}{\ell}\{2(P_{\mathsf{A}}+P_{\mathsf{B}})\}^\ell\\
&= 1-\{1-2(P_{\mathsf{A}}+P_{\mathsf{B}})\}^L.
\end{aligned} \tag{14}$$

Here, the expression is transformed by noting that, for a fixed $\ell>0$, there are $\binom{L}{\ell}$ subsets $\mathcal{S}\subseteq\mathcal{L}$ such that $|\mathcal{S}|=\ell$, and by using the assumption that the SNRs at the antennas are i.i.d.

Using $\Pr\{\Upsilon\}$, we obtain $D(2,1,L)$ as given in (13).

b) $D(2,2,L)$: We derive the probability that the reference packet is successfully decoded in the second threshold-based decoding step. This probability is expressed as

$$\begin{aligned}
D(2,2,L) &= \frac{1}{2}\big(\Pr\{\Delta\}+\Pr\{\Xi\}\big)\\
&= \frac{1}{2}\Big\{1+2(1-2P_{\mathsf{A}}-2P_{\mathsf{B}}-2P_{\mathsf{C}}-P_{\mathsf{D}})^L\\
&\quad -2(1-2P_{\mathsf{A}}-P_{\mathsf{B}}-2P_{\mathsf{C}}-P_{\mathsf{D}})^L-(1-2P_{\mathsf{A}}-2P_{\mathsf{B}})^L\Big\}.
\end{aligned} \tag{15}$$

Here, $\Pr\{\Delta\}$ denotes the probability that two packets are decoded at the same antenna in $\mathcal{L}$. In contrast, $\Pr\{\Xi\}$ represents the probability that the packets are successfully decoded across at least two distinct antennas. This probability is derived by considering all possible antenna combinations for two scenarios: (1) P_l and P_k are decoded independently at distinct antennas, respectively; and (2) the initial decoding of P_l (P_k) in an antenna set $\mathcal{L}_1\subset\mathcal{L}$ with $|\mathcal{L}_1|=\ell_1$ and its subsequent cancellation via inter-antenna SIC enables the decoding of P_k (P_l) in a distinct set $\mathcal{L}_2\subset\mathcal{L}\setminus\mathcal{L}_1$ with $|\mathcal{L}_2|=\ell_2$. Moreover, the reference packet decoded in the second step is either P_k or P_l with probability $1/2$.

In the following, we derive these two probabilities. We first consider the case where both packets are decoded at a

single antenna $i \in \mathcal{L}$. The probability of this event is given by $\Pr\{\Delta_i\} = \Pr\{\mathsf{A}_i\} + \Pr\{\breve{\mathsf{A}}_i\} = 2P_{\mathsf{A}}$. Consequently, the probability that P_k and P_l are successfully decoded at any antenna in $\mathcal{L}$ is $\Pr\{\Delta\} = \Pr\{\bigcup_{i\in\mathcal{L}} \Delta_i\}$. Using the inclusion-exclusion principle (11) and the binomial theorem, we obtain

$$\Pr\{\Delta\} = \Pr\{\bigcup_{i\in\mathcal{L}} \Delta_i\} = 1 - (1-2P_{\mathsf{A}})^L.$$

The derivation is analogous to (14) and is omitted for brevity.

Next, we derive the probability $\Pr\{\Xi\}$ that the two packets are successfully decoded at different antennas. To this end, consider the event $\Xi_{ijp}^{\mathsf{P}_l\to\mathsf{P}_k}$, which represents the situation where only P_l is independently decodable at Ant# i, P_k is decodable or in a PD state at Ant# j, and P_l and P_k cannot be simultaneously decoded at Ant# p.

This configuration implies that (1) P_l and P_k are independently and successfully decoded at Ant# i and Ant# j, respectively, and (2) the initial decoding of P_l at Ant# i, followed by its subsequent removal at Ant# j through inter-antenna SIC, enables the exclusive decoding of P_k.

For distinct i, j, and p, this event is expressed as follows:

$$\Xi_{ijp}^{\mathsf{P}_l\to\mathsf{P}_k} = \breve{\mathsf{B}}_i \cap \{\mathsf{B}_j \cup \mathsf{C}_j \cup \breve{\mathsf{C}}_j \cup \mathsf{D}_j\} \cap \{\overline{\mathsf{A}_p \cup \breve{\mathsf{A}}_p}\}. \quad (16)$$

Define $\mathcal{L}_1 \subset \mathcal{L}$ as the subset of antenna elements where only P_l is independently decodable. Define $\mathcal{L}_2 \subseteq \mathcal{L}\backslash\mathcal{L}_1$ as the subset where P_k is decodable or in PD state within each antenna element. Let $\mathcal{L}_3 = \mathcal{L}\setminus(\mathcal{L}_1\cup\mathcal{L}_2)$ denote the subset comprising the remaining antenna elements at which P_l and P_k cannot be simultaneously decoded. Let $\ell_n = |\mathcal{L}_n|$, $n = 1, 2, 3$. Note that $\mathcal{L}_1\cup\mathcal{L}_2\cup\mathcal{L}_3 = \mathcal{L}$ and $\mathcal{L}_{n_1}\cap\mathcal{L}_{n_2} = \emptyset$ for $n_1 \neq n_2$.

For any $i \in \mathcal{L}_1$, $j \in \mathcal{L}_2$, and $p \in \mathcal{L}_3$, the intersection of sets (16) is written as

$$\begin{aligned}
&\Pr\{\bigcap_{\substack{i\in\mathcal{L}_1, j\in\mathcal{L}_2,\\ p\in\mathcal{L}\setminus(\mathcal{L}_1\cup\mathcal{L}_2)}} \Xi_{ijp}^{\mathsf{P}_l\to\mathsf{P}_k}\}\\
&= \bigcap_{i\in\mathcal{L}_1}\Pr\{\breve{\mathsf{B}}_i\}\bigcap_{j\in\mathcal{L}_2}\Pr\{\mathsf{B}_j\cup\mathsf{C}_j\cup\breve{\mathsf{C}}_j\cup\mathsf{D}_j\}\bigcap_{p\in\mathcal{L}\setminus(\mathcal{L}_1\cup\mathcal{L}_2)}\Pr\{\overline{\mathsf{A}_p\cup\breve{\mathsf{A}}_p}\}\\
&= P_{\mathsf{B}}^{\ell_1}(P_{\mathsf{B}} + 2P_{\mathsf{C}} + P_{\mathsf{D}})^{\ell_2}(1-2P_{\mathsf{A}})^{L-\ell_1-\ell_2}\\
&\triangleq \Pr\{\Xi_{\mathcal{L}_1\mathcal{L}_2}^{\mathsf{P}_l\to\mathsf{P}_k}\}.
\end{aligned}$$

Using the inclusion-exclusion principle (11), we have

$$\begin{aligned}
\Pr\{\Xi^{\mathsf{P}_l\to\mathsf{P}_k}\} &= \Pr\{\bigcup_{\substack{i,j,p\in\mathcal{L},\\ i\neq j\neq p}} \Xi_{ijp}^{\mathsf{P}_l\to\mathsf{P}_k}\}\\
&= \sum_{\substack{\mathcal{L}_1\subset\mathcal{L}\\ \mathcal{L}_1\neq\emptyset}}\sum_{\substack{\mathcal{L}_2\subseteq\mathcal{L}\setminus\mathcal{L}_1\\ \mathcal{L}_2\neq\emptyset}} (-1)^{|\mathcal{L}_1|+|\mathcal{L}_2|+2}\Pr\{\bigcap_{\substack{i\in\mathcal{L}_1, j\in\mathcal{L}_2,\\ p\in\mathcal{L}\setminus(\mathcal{L}_1\cup\mathcal{L}_2)}} \Xi_{ijp}^{\mathsf{P}_l\to\mathsf{P}_k}\}\\
&= \sum_{\ell_1=1}^{L-1}\sum_{\ell_2=1}^{L-\ell_1}(-1)^{\ell_1+\ell_2}\sum_{\substack{\mathcal{L}_1\subset\mathcal{L}\\ |\mathcal{L}_1|=\ell_1}}\sum_{\substack{\mathcal{L}_2\subseteq\mathcal{L}\setminus\mathcal{L}_1\\ |\mathcal{L}_2|=\ell_2}}\Pr\{\bigcap_{\substack{i\in\mathcal{L}_1, j\in\mathcal{L}_2,\\ p\in\mathcal{L}\setminus(\mathcal{L}_1\cup\mathcal{L}_2)}} \Xi_{ijp}^{\mathsf{P}_l\to\mathsf{P}_k}\}\\
&= \sum_{\ell_1=1}^{L-1}\sum_{\ell_2=1}^{L-\ell_1}(-1)^{\ell_1+\ell_2}\binom{L}{\ell_1}\binom{L-\ell_1}{\ell_2}\Pr\{\Xi_{\mathcal{L}_1\mathcal{L}_2}^{\mathsf{P}_l\to\mathsf{P}_k}\}.
\end{aligned}$$

The term $\binom{L}{\ell_1}\binom{L-\ell_1}{\ell_2}$ accounts for the number of distinct antenna pairs $(\mathcal{L}_1, \mathcal{L}_2)$ of sizes ℓ_1 and ℓ_2, respectively, such that $\mathcal{L}_1$ and $\mathcal{L}_2$ are disjoint subsets of $\mathcal{L}$.

The symmetrical property of the SNRs for P_k and P_l ensures that $\Pr\{\Xi^{\mathsf{P}_l\leftarrow\mathsf{P}_k}\} = \Pr\{\Xi^{\mathsf{P}_l\to\mathsf{P}_k}\}$. The probability that P_k and P_l are independently and successfully decoded at each antenna element in their respective $\mathcal{L}_1$ and $\mathcal{L}_2$ is $\Pr\{\Xi_{\mathcal{L}_1\mathcal{L}_2}^{\mathsf{P}_l\to\mathsf{P}_k}\cap\Xi_{\mathcal{L}_1\mathcal{L}_2}^{\mathsf{P}_l\leftarrow\mathsf{P}_k}\} = P_{\mathsf{B}}^{\ell_1+\ell_2}(1-2P_{\mathsf{A}})^{L-\ell_1-\ell_2}$.

Therefore, we have

$$\begin{aligned}
\Pr\{\Xi\} &= \Pr\{\Xi^{\mathsf{P}_l\to\mathsf{P}_k}\}+\Pr\{\Xi^{\mathsf{P}_l\leftarrow\mathsf{P}_k}\}-\Pr\{\Xi^{\mathsf{P}_l\to\mathsf{P}_k}\cap\Xi^{\mathsf{P}_l\leftarrow\mathsf{P}_k}\}\\
&= \sum_{\ell_1=1}^{L-1}\sum_{\ell_2=1}^{L-\ell_1}(-1)^{\ell_1+\ell_2}\binom{L}{\ell_1}\binom{L-\ell_1}{\ell_2}\xi_{\Xi}(\ell_1,\ell_2),
\end{aligned}$$

where

$$\xi_{\Xi}(\ell_1,\ell_2) = (2P_{\mathsf{B}}^{\ell_1}(P_{\mathsf{B}}+2P_{\mathsf{C}}+P_{\mathsf{D}})^{\ell_2} - P_{\mathsf{B}}^{\ell_1+\ell_2})(1-2P_{\mathsf{A}})^{L-\ell_1-\ell_2}.$$

Define $\ell_3 = \ell_1 + \ell_2$ with $\ell_3 \in \{2, 3, \ldots, L\}$. It follows that $\binom{L}{\ell_1}\binom{L-\ell_1}{\ell_2} = \binom{L}{\ell_3}\binom{\ell_3}{\ell_1}$. Thus, $\Pr\{\Xi\}$ can be rewritten as

$$\begin{aligned}
\Pr\{\Xi\} &= \sum_{\ell_1=1}^{L-1}\sum_{\ell_3=\ell_1+1}^{L}(-1)^{\ell_3}\binom{L}{\ell_3}\binom{\ell_3}{\ell_1}\xi_{\Xi}(\ell_1,\ell_3-\ell_1)\\
&= \sum_{\ell_3=2}^{L}\sum_{\ell_1=1}^{\ell_3-1}(-1)^{\ell_3}\binom{L}{\ell_3}\binom{\ell_3}{\ell_1}\\
&\quad\cdot(2P_{\mathsf{B}}^{\ell_1}(P_{\mathsf{B}}+2P_{\mathsf{C}}+P_{\mathsf{D}})^{\ell_3-\ell_1} - P_{\mathsf{B}}^{\ell_3})(1-2P_{\mathsf{A}})^{L-\ell_3}.
\end{aligned}$$

Finally, by applying the binomial theorem, we obtain

$$\begin{aligned}
\Pr\{\Xi\} = &2(1-2P_{\mathsf{A}}-2P_{\mathsf{B}}-2P_{\mathsf{C}}-P_{\mathsf{D}})^L - 2(1-2P_{\mathsf{A}}-P_{\mathsf{B}}-2P_{\mathsf{C}}-P_{\mathsf{D}})^L\\
&-(1-2P_{\mathsf{A}}-2P_{\mathsf{B}})^L+(1-2P_{\mathsf{A}})^L.
\end{aligned}$$

With $\Pr\{\Delta\}$ and $\Pr\{\Xi\}$, we obtain $D(2,2,L)$ in (15).

Remark 1. *For the single-antenna case $L = 1$, $R \geq 1$, and $m \leq 2$, the average decoding error probability $\bar{\epsilon}_{m,1}$ in* (10) *reduces to the result in [6]. A contribution of this work is the derivation of the exact average decoding error probability for the broader range $R > 0$ with $m \leq 2$, by characterizing the integration regions for $0 < R < 1$ as detailed in Sec. III-C.* □

B. Closed-Form Approximation for $m \geq 3$

In this subsection, we consider the case of $m \geq 3$. In multi-antenna reception, a packet decoded at one antenna is removed from all antennas through inter-antenna SIC. Consequently, the decoding condition at each SIC step depends on the outcomes of the preceding SIC steps, resulting in statistical dependence among the decoding events. As shown in the previous subsection, the exact expressions for $m \leq 2$ are obtained by exhaustively enumerating the possible decoding events and applying the inclusion-exclusion principle. However, the number of such events grows rapidly with m, making this approach analytically intractable for higher-order collisions.

To derive a closed-form approximation for $D(m, L)$, let $\mathcal{A}_i$ denote the event that the ith non-reference packet is decoded at the first SIC step, and let $\mathcal{B}_i$ denote the event that the reference packet is eventually decoded after $\mathcal{A}_i$ occurs. Conditioning on the packet decoded at the first SIC step yields

$$D(m, L) = D(m, 1, L) + (m-1)D(m, 1, L)\Pr(\mathcal{B}_i \mid \mathcal{A}_i). \quad (17)$$

The first term on the right-hand side represents the probability that the reference packet is decoded at the first SIC step. The second term represents the probability that one of the $m-1$ non-reference packets is decoded first and canceled, after which the reference packet is eventually decoded from the residual $(m-1)$-packet collision.

To obtain a tractable approximation, after each successful cancellation, the residual collision is modeled as an independent collision with the same statistical characteristics as a newly generated collision consisting of the remaining packets. Hence,

$$\Pr(\mathcal{B}_i \mid \mathcal{A}_i) \approx \widetilde{D}(m-1, L), \tag{18}$$

where $\widetilde{D}(m, L)$ denotes the approximate probability that the reference packet is eventually decoded from an m-packet collision. Substituting (18) into (17) gives

$$\begin{aligned} D(m, L) &\approx \widetilde{D}(m, L) \\ &= D(m, 1, L)\left(1 + (m-1)\widetilde{D}(m-1, L)\right), \ m \geq 3. \end{aligned}$$

Accordingly, $\widetilde{D}(m, L)$ is recursively defined as

$$\widetilde{D}(m, L) = \begin{cases} D(2, L), & m = 2, \\ D(m, 1, L)\Big(1 + (m-1)\widetilde{D}(m-1, L)\Big), & m \geq 3, \end{cases} \tag{19}$$

where the exact two-packet decoding probability $D(2, L) = D(2, 1, L) + D(2, 2, L)$ is obtained from (13) and (15). By recursively expanding (19), $\widetilde{D}(m, L)$ can equivalently be expressed in the following closed form:

$$\begin{aligned} \widetilde{D}(m, L) = &\sum_{j=3}^{m} \frac{(m-1)!}{(j-1)!} \prod_{r=j}^{m} D(r, 1, L) \\ &+ (m-1)! D(2, L) \prod_{r=3}^{m} D(r, 1, L), \ \ m \geq 3. \end{aligned} \tag{20}$$

The term $D(r, 1, L)$ denotes the first-step decoding probability for an arbitrary collision size r. Since its exact expression is derived in Appendix II, the approximation above can be evaluated over the entire range $R > 0$.

Finally, for $m \geq 3$, the average decoding error probability is approximated as

$$\bar{\epsilon}_{m,L} \approx 1 - \widetilde{D}(m, L). \tag{21}$$

Remark 2. *Focusing first on the case $m = 2$, the average decoding error probability $\bar{\epsilon}_{2,L}$ of MRC-SIC in [23, Theorem 3] is obtained using an approximation whose accuracy improves in the large-L regime. In contrast, the SC-SIC analysis yields an exact closed-form expression for $\bar{\epsilon}_{2,L}$ that is valid for any finite L. For $m \geq 3$, the proposed approximation explicitly accounts for decoding of the reference packet after preceding interference cancellations, while the normal approximation in MRC-SIC [23, Theorem 3] does not explicitly represent the subsequent SIC steps. As shown in Table II, both the exact result for $m = 2$ and the approximation for $m \geq 3$ agree closely with the corresponding SC-SIC simulation results over the considered finite values of L, whereas the MRC-SIC approximations show a noticeable mismatch.* □

TABLE II

AVERAGE DECODING ERROR PROBABILITY $\bar{\epsilon}_{m,L}$ $(m = 1, 2, 3, 4)$ WITHIN A SLOT. "Gap" DENOTES THE ABSOLUTE DIFFERENCE $|\text{Sim.} - \text{Theo.}|$. SIMULATION RESULTS REPORTED AS $< 10^{-6}$ INDICATE THAT NO DECODING ERROR WAS OBSERVED OVER 10^6 MONTE CARLO TRIALS. $(\eta_0 = 10, \bar{\gamma} = 10\,\text{dB})$.

		MRC-SIC [23, Theorem 3]			SC-SIC of (10) or (21)		
m	L	Sim.	Theo.	Gap	Sim.	Theo.	Gap
1	4	1.88×10^{-2}	1.90×10^{-2}	1.7×10^{-4}	1.60×10^{-1}	1.60×10^{-1}	1.0×10^{-4}
	8	1.00×10^{-5}	1.00×10^{-5}	$< 10^{-6}$	2.53×10^{-2}	2.55×10^{-2}	2.0×10^{-4}
	16	$< 10^{-6}$	$< 10^{-6}$	$< 10^{-6}$	7.00×10^{-4}	7.20×10^{-4}	1.8×10^{-5}
	32	$< 10^{-6}$	$< 10^{-6}$	$< 10^{-6}$	$< 10^{-6}$	$< 10^{-6}$	$< 10^{-6}$
	64	$< 10^{-6}$	$< 10^{-6}$	$< 10^{-6}$	$< 10^{-6}$	$< 10^{-6}$	$< 10^{-6}$
2	4	6.16×10^{-1}	7.32×10^{-1}	1.2×10^{-1}	7.89×10^{-1}	7.89×10^{-1}	5.1×10^{-5}
	8	3.68×10^{-1}	4.81×10^{-1}	1.1×10^{-1}	5.84×10^{-1}	5.84×10^{-1}	9.7×10^{-5}
	16	1.41×10^{-1}	2.07×10^{-1}	6.6×10^{-2}	3.31×10^{-1}	3.31×10^{-1}	1.7×10^{-4}
	32	2.25×10^{-2}	3.84×10^{-2}	1.6×10^{-2}	1.10×10^{-1}	1.09×10^{-1}	3.9×10^{-4}
	64	7.00×10^{-4}	1.30×10^{-3}	6.0×10^{-4}	1.19×10^{-2}	1.19×10^{-2}	3.1×10^{-5}
3	4	9.38×10^{-1}	9.20×10^{-1}	1.8×10^{-2}	9.85×10^{-1}	9.83×10^{-1}	1.7×10^{-3}
	8	7.53×10^{-1}	8.31×10^{-1}	7.8×10^{-2}	9.60×10^{-1}	9.57×10^{-1}	2.8×10^{-3}
	16	3.82×10^{-1}	6.20×10^{-1}	2.4×10^{-1}	8.97×10^{-1}	8.94×10^{-1}	3.5×10^{-3}
	32	6.85×10^{-2}	1.92×10^{-1}	1.2×10^{-1}	7.67×10^{-1}	7.64×10^{-1}	2.7×10^{-3}
	64	7.93×10^{-4}	1.66×10^{-3}	8.6×10^{-4}	5.61×10^{-1}	5.60×10^{-1}	1.1×10^{-3}
4	4	9.92×10^{-1}	9.68×10^{-1}	2.4×10^{-2}	9.99×10^{-1}	9.99×10^{-1}	2.1×10^{-5}
	8	9.43×10^{-1}	9.23×10^{-1}	2.0×10^{-2}	9.98×10^{-1}	9.98×10^{-1}	6.9×10^{-5}
	16	6.76×10^{-1}	8.05×10^{-1}	1.3×10^{-1}	9.94×10^{-1}	9.94×10^{-1}	2.6×10^{-4}
	32	1.61×10^{-1}	4.53×10^{-1}	2.9×10^{-1}	9.86×10^{-1}	9.85×10^{-1}	4.8×10^{-4}
	64	3.39×10^{-2}	2.32×10^{-2}	1.1×10^{-2}	9.62×10^{-1}	9.60×10^{-1}	1.1×10^{-3}

C. High-SNR Behavior for Finite L ($\bar{\gamma} \to \infty$)

In this subsection, we derive the high-SNR limit of the average decoding error probability $\bar{\epsilon}_{m,L}$ for a fixed finite number of receive antennas L. The objective is to identify the conditions under which the decoding error probability vanishes as $\bar{\gamma} \to \infty$ and those under which a nonzero error floor persists because of residual multiuser interference. We also characterize the effects of the collision size m and the number of receive antennas L on the high-SNR performance.

For a singleton slot, the average decoding error probability vanishes in the high-SNR regime, i.e.,

$$\lim_{\bar{\gamma} \to \infty} \bar{\epsilon}_{1,L} = 0. \tag{22}$$

This result follows directly from (12).

We next consider a residual m-packet collision with $m \geq 2$. Let $\gamma_{1,i} > \gamma_{2,i} > \cdots > \gamma_{m,i}$ denote the ordered received SNRs at the ith receive antenna. The SINR of the strongest packet satisfies

$$\eta_{1,i} = \frac{\gamma_{1,i}}{1 + \sum_{k=2}^{m} \gamma_{k,i}} > \frac{\gamma_{1,i}}{1 + (m-1)\gamma_{1,i}}. \tag{23}$$

As $\bar{\gamma} \to \infty$, the lower bound converges to

$$\lim_{\bar{\gamma} \to \infty} \frac{\gamma_{1,i}}{1 + (m-1)\gamma_{1,i}} = \frac{1}{m-1}. \tag{24}$$

Therefore, if $\eta_0 \leq 1/(m-1)$, the strongest packet is decodable almost surely in the high-SNR regime. After its cancellation, the same argument applies recursively to the residual collision. Consequently, all m packets, including the reference packet, are eventually decoded, and no high-SNR error floor remains.

Conversely, if $\eta_0 > 1/(m-1)$, in the high-SNR limit, the SINR at the equal-power point is $\eta_{j,i} = 1/(m-1) < \eta_0$ for all $j = 1, \ldots, m$ and $i = 1, \ldots, L$. By continuity, there exists a neighborhood in which $\eta_{j,i} < \eta_0$ for all j and i, preventing SIC. Since this neighborhood has positive probability under continuous fading, a collision remains undecodable with positive probability as $\bar{\gamma} \to \infty$, resulting in a nonzero error floor.

For $m \geq 2$, the high-SNR behavior is characterized as

$$\lim_{\bar{\gamma}\to\infty} \bar{\epsilon}_{m,L} = \begin{cases} 0, & 0 < \eta_0 \leq \dfrac{1}{m-1}, \\ \bar{\epsilon}_{m,L}^{\text{floor}}(\eta_0) > 0, & \eta_0 > \dfrac{1}{m-1}. \end{cases} \tag{25}$$

For $m = 2$, the nonzero high-SNR error floor is obtained exactly, whereas for $m \geq 3$, it is approximated by taking the high-SNR limit of the proposed approximation:

$$\bar{\epsilon}_{2,L}^{\text{floor}}(\eta_0) = \left(\frac{\eta_0 - 1}{\eta_0 + 1}\right)^L, \tag{26}$$

$$\begin{aligned} \bar{\epsilon}_{m,L}^{\text{floor}}(\eta_0) &\approx \widetilde{\bar{\epsilon}}_{m,L}^{\text{floor}}(\eta_0) \\ &= 1 - \frac{1}{m}\left[\sum_{j=3}^{m}\prod_{r=j}^{m}\Phi_{r,L}(\eta_0) + 2\prod_{r=2}^{m}\Phi_{r,L}(\eta_0)\right], \end{aligned} \tag{27}$$

where

$$\Phi_{r,L}(\eta_0) = 1 - \left[1 - \sum_{\substack{1\leq s\leq r \\ 1-(s-1)\eta_0>0}} (-1)^{s+1}\binom{r}{s}\left(\frac{1-(s-1)\eta_0}{1+\eta_0}\right)^{r-1}\right]^L. \tag{28}$$

The exact result in (26) and the approximate result in (27) are derived in Appendix III.

V. Density Evolution with L-Antenna Reception

In this section, building on the DE framework for IR-SA [6]–[9], we incorporate the L-antenna decoding probabilities derived in Section IV into the SN update and characterize the asymptotic performance ($M, N_{\text{T}} \to \infty$) of multi-antenna IR-SA.

We refer to the DNs (SNs) collectively as the DN (SN) group. In the protograph, let ${}^{\ell}p$ denote the probability that an edge incident on the DN group carries an erasure message toward the SN group at iteration ℓ. Similarly, let ${}^{\ell}q$ denote the probability that an edge incident on the SN group carries an erasure message toward the DN groups.

First, consider the DN group with degree distribution $\lambda(x)$. At iteration ℓ, an edge incident to a degree-l DN is removed if at least one of the other $l-1$ edges has been successfully resolved. Therefore, given the device activation probability π, the average probability that a DN-to-SN message remains erased is given by [7]–[9]

$${}^{\ell}p = \pi \textstyle\sum_{l=1}^{l_{\max}} \lambda_l \cdot {}^{(\ell-1)}q^{l-1} = \pi\lambda({}^{(\ell-1)}q). \tag{29}$$

Second, consider the SN group, which has degree distribution $\rho(x)$. The average probability that an SN-to-DN message remains erased at iteration ℓ is given by ${}^{\ell}q = \sum_{r=1}^{\infty}\rho_r \cdot {}^{\ell}q^{(r)}$, where ρ_r is the fraction of edges connected to an SN of degree-r (see (3)), and ${}^{\ell}q^{(r)}$ denotes the conditional erasure probability of an outgoing message from a degree-r SN. Following [7], it is given by

$${}^{\ell}q^{(r)} = 1 - \textstyle\sum_{m=1}^{r}\binom{r-1}{m-1}(1-{}^{\ell}p)^{r-m}({}^{\ell}p)^{m-1}(1-\bar{\epsilon}_{m,L}). \tag{30}$$

Here, $(1-\bar{\epsilon}_{m,L})$ denotes the probability that the replica associated with an outgoing edge is successfully decoded when the residual degree of the SN is m. Specifically, among the other $r-1$ replicas in a degree-r slot, $r-m$ replicas have been removed through inter-slot SIC, leaving the reference replica together with $m-1$ interfering replicas. The resulting m-packet collision is then processed by the L-antenna receiver, whose decoding operation is fully encapsulated by $1-\bar{\epsilon}_{m,L}$. The corresponding $\bar{\epsilon}_{m,L}$ is derived in Section IV.

Averaging over the SN degree distribution and simplifying as in [6], [7], we obtain

$${}^{\ell}q = 1 - e^{-\alpha\bar{d}\,{}^{\ell}p}\textstyle\sum_{m=0}^{\infty}\frac{(\alpha\bar{d}\,{}^{\ell}p)^m}{m!}(1-\bar{\epsilon}_{m+1,L}). \tag{31}$$

From (2), we have $\alpha\bar{d}x = -\ln\rho(1-x)$, which allows ${}^{\ell}q$ to be expressed as a function of $(1-{}^{\ell}p)$:

$${}^{\ell}q = 1 - \chi_L(1-{}^{\ell}p), \tag{32}$$

where the L-antenna SN decoding function $\chi_L(\cdot)$, which encapsulates the effect of multi-antenna reception through $1-\bar{\epsilon}_{m,L}$, is defined as

$$\chi_L(1-x) \triangleq e^{-\alpha\bar{d}x}\sum_{m=0}^{\infty}\frac{(\alpha\bar{d}x)^m}{m!}(1-\bar{\epsilon}_{m+1,L}). \tag{33}$$

Thus, all effects of L-antenna reception are encapsulated within the SN decoding function $\chi_L(\cdot)$, while the graph-level DE structure retains the same form as in the single-antenna case.

Substituting (29) into (31) and using $G = \pi\alpha$, the DE recursion becomes, with ${}^{(\ell=0)}q = 1$,

$${}^{\ell}q = 1 - e^{-G\bar{d}\lambda({}^{(\ell-1)}q)}\sum_{m=0}^{\infty}\frac{(G\bar{d}\lambda({}^{(\ell-1)}q))^m}{m!}(1-\bar{\epsilon}_{m+1,L}). \tag{34}$$

Let ${}^{(\infty)}q(G,\bar{d},L,\bar{\gamma},\eta_0) = \lim_{\ell}{}^{(\ell)}q$ denote the limiting erasure probability of the DE recursion. The corresponding asymptotic PLR is [6], [9]

$$\begin{aligned} \text{PLR}(G,\bar{d},L,\bar{\gamma},\eta_0) &= \Lambda({}^{(\infty)}q(G,\bar{d},L,\bar{\gamma},\eta_0)) \\ &= \sum_{l=1}^{l_{\max}}\Lambda_l\left({}^{(\infty)}q(G,\bar{d},L,\bar{\gamma},\eta_0)\right)^l. \end{aligned} \tag{35}$$

For a target PLR^*, the asymptotic iterative decoding threshold, also referred to as the BP decoding threshold, is defined as the supremum offered traffic [6], [9] for which the asymptotic PLR does not exceed PLR^*

$$G^{\text{BP,blk}} = \sup\{G : \text{PLR}(G,\bar{d},L,\bar{\gamma},\eta_0) \leq \text{PLR}^*\}. \tag{36}$$

Under the κ-MaxDecoding constraint defined in Remark 3, the summation in (34) is restricted to $m = 0,\ldots,\kappa-1$, since the corresponding residual degree is $m+1 \leq \kappa$.

Remark 3 (κ-MaxDecoding)**.** *In* (34)*, the summation extends to infinity due to the unbounded support of the Poisson distribution. However, practical systems typically restrict SIC to low-order collisions because of interference, error propagation, and computational complexity [30]. Accordingly, we adopt κ-MaxDecoding, in which intra-slot SIC in Section III-A is performed only when the number of residual packets in a slot does not exceed κ. Hence, a slot initially containing an m-packet collision with $m > \kappa$ may still be decoded after inter-slot SIC if its residual degree is at most κ.* □

VI. Converse Bound and MAP Threshold

Building on the EXIT-area and trial-entropy framework for single-antenna IR-SA in [7], we extend the converse-bound and MAP-threshold analyses to finite-L multi-antenna reception through the SN decoding function $\chi_L(\cdot)$ derived in Section V.

A. Converse Bound

Following the EXIT-area argument for single-antenna IR-SA in [7, Sec. III-A], a necessary condition for successful iterative decoding is that the DN and SN EXIT functions form an open decoding tunnel. Suppressing the iteration index in (29) and (32), the corresponding transfer functions for the present multi-antenna system are

$$p = \pi\lambda(q) \triangleq f_{\mathrm{d}}(q), \qquad q \in (0,1], \tag{37}$$

$$q = 1 - \chi_L(1-p) \triangleq f_{\mathrm{s}}(p), \qquad p \in (0,\pi]. \tag{38}$$

The multi-antenna receiver affects this EXIT representation only through the SN decoding function $\chi_L(\cdot)$, which incorporates the finite-L decoding success probabilities $1-\bar{\epsilon}_{m,L}$ defined in (33).

As established in [7], the open-tunnel condition requires [24]–[26]

$$A_{\mathrm{d}} + A_{\mathrm{s}} < \pi, \tag{39}$$

where the areas associated with the DN and SN EXIT functions are given by

$$\begin{aligned} A_{\mathrm{d}} &= \int_0^1 f_{\mathrm{d}}(q)\mathrm{d}q = \pi\sum_l \lambda_l \int_0^1 q^{l-1}\mathrm{d}q \\ &= \pi\sum_l \frac{\lambda_l}{l} = \frac{\pi}{\Lambda'(1)} = \frac{\pi}{\bar{d}}, \end{aligned} \tag{40}$$

$$\begin{aligned} A_{\mathrm{s}} &= \int_0^{\pi} f_{\mathrm{s}}(p)\mathrm{d}p = \int_0^{\pi} \left(1-\chi_L(1-p)\right)\mathrm{d}p \\ &= \pi - \tau_L(\pi), \end{aligned} \tag{41}$$

with

$$\tau_L(x) \triangleq \int_0^x \chi_L(1-z)\mathrm{d}z. \tag{42}$$

Under the κ-MaxDecoding constraint in Remark 3, substituting (33) into (42) yields

$$\tau_L(x) = \frac{1}{\alpha\bar{d}} \sum_{m=0}^{\kappa-1} \left(1-\bar{\epsilon}_{m+1,L}\right) \left[1 - e^{-\alpha\bar{d}x} \sum_{k=0}^{m} \frac{(\alpha\bar{d}x)^k}{k!}\right]. \tag{43}$$

Combining (39), (40), and (41), the open-tunnel condition becomes $\pi\alpha < \alpha\bar{d}\,\tau_L(\pi)$. With $G = \pi\alpha$, the boundary of this condition, $\pi\alpha = \alpha\bar{d}\,\tau_L(\pi)$, is given by

$$G^{\mathrm{C,blk}} = \sum_{m=0}^{\kappa-1} \left(1-\bar{\epsilon}_{m+1,L}\right) \left[1 - e^{-G^{\mathrm{C,blk}}\bar{d}} \sum_{k=0}^{m} \frac{(G^{\mathrm{C,blk}}\bar{d})^k}{k!}\right]. \tag{44}$$

The positive solution of (44) gives the converse bound $G^{\mathrm{C,blk}}$.

Equation (44) extends the single-antenna converse bound in [7] to finite-L multi-antenna reception through the decoding success probabilities $1-\bar{\epsilon}_{m+1,L}$. Consequently, for a fixed average repetition degree $\bar{d}$, $G^{\mathrm{C,blk}}$ is independent of the particular DN degree distribution $\Lambda(x)$. For $L=1$, (44) reduces to the single-antenna converse bound in [7].

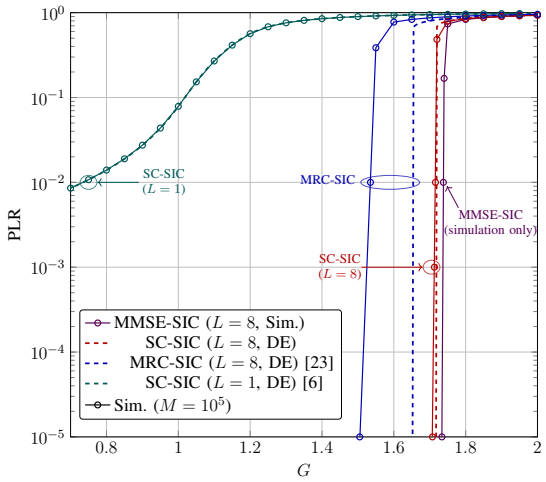


Fig. 5. PLRs vs. offered traffic G of IR-SA for the SC-SIC, the MRC-SIC [23], and the MMSE-SIC (simulation only): $\kappa = 2$, $\eta_0 = 5$, $\bar{\gamma} = 20$ dB, and $\Lambda(x) = 0.586x^2 + 0.291x^3 + 0.031x^6 + 0.022x^7 + 0.006x^8 + 0.063x^{10}$.

B. MAP Threshold

Following the trial-entropy analysis for single-antenna IR-SA, we extend the MAP-threshold characterization to finite-L multi-antenna reception through the SN decoding function $\chi_L(\cdot)$; see [7, Sec. III-B] for details of the underlying derivation.

The limiting DN-to-SN erasure probability x satisfies the fixed-point equation

$$x = \pi\lambda\left(1-\chi_L(1-x)\right), \tag{45}$$

which gives

$$\pi_L(x) = \frac{x}{\lambda\left(1-\chi_L(1-x)\right)}, \qquad x \in (0,\bar{x}], \tag{46}$$

where $\pi_L(\bar{x}) = 1$ and $0 < \bar{x} < 1$.

The corresponding trial entropy is

$$P_L(x) = \frac{x\Lambda\left(1-\chi_L(1-x)\right)}{\lambda\left(1-\chi_L(1-x)\right)} + \Lambda'(1)\left[x\chi_L(1-x) - \tau_L(x)\right], \tag{47}$$

where $\tau_L(x)$ is defined in (42).

The MAP threshold is obtained from the nonzero solution x^{MAP} of $P_L(x^{\mathrm{MAP}}) = 0$, yielding

$$G^{\mathrm{MAP,blk}} = \alpha\pi_L(x^{\mathrm{MAP}}). \tag{48}$$

For $L=1$, this MAP-threshold characterization reduces to the single-antenna result in [7].

VII. Numerical and Simulation Results

In this section, using the derived $\bar{\epsilon}_{m,L}$ and the DE equations, we present numerical results for the BP thresholds $G^{\mathrm{BP,blk}}$ in (36), along with the corresponding MAP thresholds $G^{\mathrm{MAP,blk}}$ (48) and converse bounds $G^{\mathrm{C,blk}}$ (44).

Figure 5 presents the PLR performance of the proposed SC-SIC in (35) for $L = 8$, $\kappa = 2$ at the target $\mathrm{PLR}^* = 10^{-2}$. For comparison, the corresponding performance of multi-antenna IR-SA using MRC-SIC [23] and minimum mean-square error-based SIC (MMSE-SIC) [31] is also shown. For reference, the performance of the single-antenna IR-SA ($L = 1$) [6] is also included. The same degree distribution, optimized specifically for the proposed multi-antenna IR-SA via a differential evolution algorithm [32], is applied to all schemes. Since no analytical result is available for MMSE-SIC, its performance is evaluated by simulation only, where threshold-based decoding is applied after MMSE filtering. Simulation results with a finite number of slots ($M = 10^5$) for SC-SIC closely match the asymptotic DE analysis, thereby validating our analytical framework. In contrast, the MRC-SIC approach in [23] exhibits a noticeable discrepancy between simulation and analysis. This discrepancy arises because the MRC-SIC analysis is approximate even for $m = 2$, whereas the proposed SC-SIC analysis is exact for $m \leq 2$, as confirmed by the results in Table II.

Beyond the performance analysis, it is also important to consider the computational complexity of the multi-antenna processing in each decoding attempt. MMSE-SIC requires the inversion of an $L \times L$ interference-plus-noise covariance matrix for MMSE filtering, which generally entails cubic complexity with respect to L. In contrast, SC-SIC avoids such matrix operations and relies on per-antenna SINR evaluation and branch selection, whose complexity scales linearly with L. Both schemes additionally require packet reconstruction and interference cancellation as part of the iterative SIC procedure. As shown in Fig. 5, SC-SIC achieves performance close to that of MMSE-SIC while avoiding the computationally intensive matrix inversion required for MMSE filtering.

Figure 6 compares the PLRs obtained from DE with the corresponding simulation results for SC-SIC-based IR-SA as functions of the offered traffic G, for $\kappa \in \{3, 4\}$ and $L \in \{4, 8\}$. The DE results agree well with the corresponding simulation results for all considered combinations of κ and L, demonstrating the accuracy of the proposed approximation. Compared to the exact agreement between the theoretical analysis and the simulation results for $\kappa = 2$ in Fig. 5, the slight remaining discrepancies here are attributable to the independence assumption in (18).

Figures 7(a) and 7(b) plot the BP thresholds $G^{\mathrm{BP,blk}}$ of MRC-SIC [23] and SC-SIC versus the average received SNR $\bar{\gamma}$ for $\kappa = 1$ and $\kappa = 2, 3$, respectively. For $\kappa = 1$, MRC-SIC consistently outperforms SC-SIC over the entire SNR range because it maximizes the received SNR of the desired signal. Furthermore, MRC-SIC approaches the performance of the collision channel, for which $\bar{\epsilon}_{1,1} = 0$ with a single receive antenna [9], at a lower SNR than SC-SIC.

For $\kappa \geq 2$, a different trend is observed. For $\kappa = 2$, in the low-SNR regime ($\bar{\gamma} < 12$ dB), MRC-SIC achieves higher BP thresholds owing to its combining gain, which is essential for raising the effective SNR above the decoding threshold η_0. In contrast, because SC-SIC does not exploit combining gain, the SINRs at all L antennas tend to remain below η_0, resulting in decoding failures. In the high-SNR regime ($\bar{\gamma} \geq 12$ dB), a performance crossover occurs. Although SC-SIC does not benefit from combining gain, intra-antenna SIC is likely to succeed at least one or more antennas. By contrast, MRC-SIC combines the interference received across all antennas as well, thereby limiting the effective SINR. A similar performance crossover is observed for $\kappa = 3$.

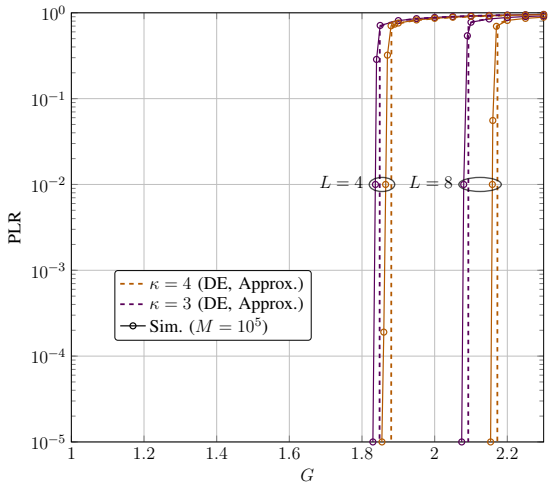
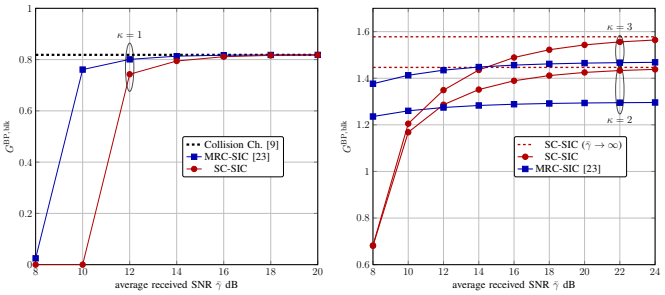


Fig. 6. PLRs vs. offered traffic G of SC-SIC-based IR-SA for $\kappa = 3$ and 4, validating the approximation. Other parameters are identical to those in Fig. 5.



(a) $\kappa = 1$, $L = 3$. **(b)** $\kappa = 2, 3$, $L = 8$.

Fig. 7. BP thresholds $G^{\mathrm{BP,blk}}$ vs. average received SNR $\bar{\gamma}$ for MRC-SIC [23] and SC-SIC: (a) $\kappa = 1$ with $L = 3$, including the collision-channel threshold ($\bar{\epsilon}_{1,1} = 0$) [9]; (b) $\kappa = 2, 3$ with $L = 8$. ($\Lambda(x) = x^3$, $\eta_0 = 10$, and $\mathrm{PLR}^* = 10^{-2}$).

Figure 7(b) also shows the high-SNR BP benchmarks obtained by substituting the high-SNR error floors of $\bar{\epsilon}_{m,L}$, derived in (26) and (27), into the DE equation (34). Unlike the case of $\kappa = 1$, for $\kappa \geq 2$, the decoding error probability does not converge to zero even as $\bar{\gamma} \to \infty$ (see (25)), and a nonzero error floor remains. For a given κ and degree distribution, these benchmarks are independent of the SNR in the high-SNR regime and can be evaluated solely from L and η_0. They therefore provide a concise characterization of the asymptotic BP performance attainable by SC-SIC with a finite number of receive antennas.

Figure 8 illustrates the case of $R < 1$ and evaluates the asymptotic offered traffic, using $\eta_0 = 0.8$ ($R = 0.848$) as an example with $\bar{\gamma} = 0$ dB for $\kappa = 2, 3$, and 4. For $\kappa = 2$, as discussed in Section III, the exact closed-form expression for the average error probability is valid over the entire range

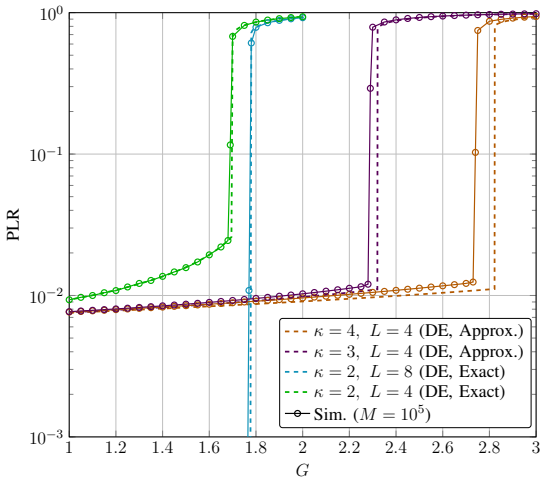


Fig. 8. PLRs vs. offered traffic G of IR-SA for SC-SIC under $R < 1$: $\kappa = 2, 3, 4$, $\bar{\gamma} = 0$ dB, $\eta_0 = 0.8$, and $\Lambda(x) = 0.760x^2 + 0.069x^3 + 0.040x^6 + 0.076x^7 + 0.033x^8 + 0.022x^{10}$.

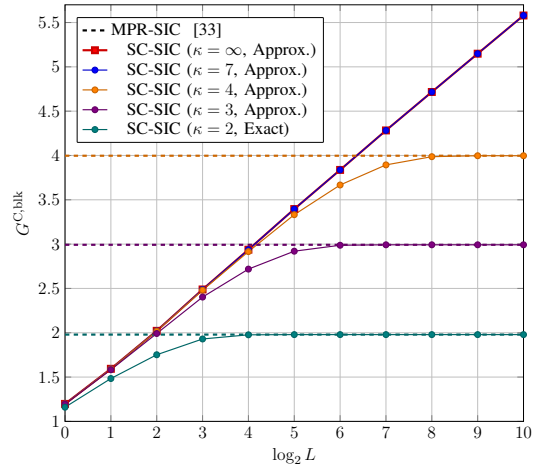


Fig. 9. Converse bounds versus $\log_2 L$ for SC-SIC with $\kappa = 2, 3, 4, 7$, and ∞, together with the corresponding κ-MPR-SIC converse bounds [33]; $\eta_0 = 5$, $\bar{\gamma} = 20$ dB, and $d = 3$.

$R > 0$, and the corresponding asymptotic results show strong agreement with the simulation results for $L = 4$ and $L = 8$.

For $\kappa = 3$ and $\kappa = 4$, the asymptotic results based on the approximation agree reasonably well with the simulations. However, slightly larger discrepancies are observed compared with the $R > 1$ case in Fig. 6, which employs the same formulation. This discrepancy arises because the approximation assumes that one of the $m - 1$ non-reference packets is decoded and canceled first, after which the reference packet is decoded from the residual $(m-1)$-packet collision. When R is sufficiently large, the exactly calculated $D(m, 1, L)$ dominates, yielding high accuracy. Conversely, as R decreases, the contribution of later SIC steps becomes more significant, slightly reducing the approximation accuracy. These results demonstrate that the proposed analytical framework remains applicable to the entire range $R > 0$, enabling the derivation of MAP thresholds and converse bounds for $R < 1$ in the same manner as for $R \geq 1$.

Figure 9 illustrates the converse bound $G^{\mathrm{C,blk}}$ derived from (44) as a function of the logarithm of the number of receive antennas $\log_2 L$, alongside the corresponding κ-multi-packet reception (MPR)-SIC converse bounds derived in [33] for comparison. In κ-MPR-SIC, an AP is assumed to successfully decode all packets in a slot containing at most κ packets, i.e., $\bar{\epsilon}_{m,1} = 0$ for $m \leq \kappa$. For each finite κ, the SC-SIC converse bound monotonically converges to the corresponding κ-MPR-SIC converse bound as L increases, becoming nearly indistinguishable from the corresponding limits at $L \approx 16$, 64, and 256 for $\kappa = 2$, 3, and 4, respectively. This demonstrates that spatial diversity sufficiently reduces $\bar{\epsilon}_{m,L}$, enabling finite-antenna SC-SIC to approach the ideal MPR collision-resolution capability. Furthermore, the two curves for $\kappa = 7$ and $\kappa = \infty$ are nearly indistinguishable for $L \leq 1024$, implying that accounting for collisions involving eight or more packets provides only a negligible additional increase in the converse bound.

Table III presents the MAP thresholds $G^{\mathrm{MAP,blk}}$ and converse bounds $G^{\mathrm{C,blk}}$ for $\kappa = 2$ and 3, together with the BP thresholds $G^{\dagger\mathrm{BP,Conv}}$ of the spatially coupled scheme

TABLE III
BP/MAP THRESHOLDS AND CONVERSE BOUNDS FOR $\kappa = 2$ AND 3. ($\eta_0 = 5$, $\bar{\gamma} = 20$ dB, $\alpha = 4$, $L_{\mathrm{sc}} = 100$, $\mathrm{PLR}^* = 10^{-2}$, and $d = 3$).

	$\kappa = 2$			$\kappa = 3$		
L	$G^{\dagger\mathrm{BP,Conv}}$	$G^{\mathrm{MAP,blk}}$	$G^{\mathrm{C,blk}}$	$G^{\dagger\mathrm{BP,Conv}}$	$G^{\mathrm{MAP,blk}}$	$G^{\mathrm{C,blk}}$
1 [7]	1.131	1.132	1.160	1.160	1.160	1.194
4	1.746	1.747	1.752	1.985	1.985	1.991
8	1.927	1.928	1.930	2.400	2.400	2.403
16	1.974	1.974	1.977	2.716	2.716	2.718
∞	1.976	1.977	1.979	2.989	2.991	2.992

extended from the single-antenna case [7], [8]. The MAP thresholds and converse bounds increase with L, demonstrating the gain provided by spatial diversity. Moreover, the MAP thresholds closely track the corresponding converse bounds for all considered values of L and κ. The resulting converse bounds are tight because the two EXIT functions in (37) and (38) closely match each other.

In the spatially coupled scheme, a superframe consists of $L_{\mathrm{sc}} + d - 1$ frames, each containing M slots. Details of the spatially coupled construction are omitted due to space limitations and can be found in [7], [8]. The numerically obtained BP thresholds nearly coincide with the corresponding MAP thresholds. This behavior is known as the threshold saturation effect in the context of LDPC codes [34]. For both $\kappa = 2$ and $\kappa = 3$, threshold saturation is observed for all considered numbers of receive antennas.

VIII. CONCLUSION

This paper developed an analytical framework for IR-SA with multi-antenna reception over Rayleigh block-fading channels. We derived exact closed-form expressions for the average decoding error probability $\bar{\epsilon}_{m,L}$ under SC-SIC for $m \leq 2$ and developed a tractable approximation for $m \geq 3$. These results enabled a unified DE-based analysis that incorporates multi-antenna reception while preserving the underlying bipartite graph structure. We further characterized the BP and MAP thresholds and derived a converse bound for the considered system. Numerical results showed that spatial coupling drives

the BP threshold toward the MAP threshold and the corresponding converse bound, demonstrating threshold saturation. The comparison with MRC-SIC further revealed that the relative advantage of the two multi-antenna decoding strategies depends on the operating SNR and the decoding threshold.

Deriving an exact characterization for higher-order collisions remains an important direction for future work. Other directions include investigating the impact of spatial correlation among antenna elements on the decoding error probability and threshold performance, and extending the analytical framework to other combining schemes and more general fading models.

APPENDIX I
CLOSED-FORM PROBABILITY EXPRESSIONS

A. $R \geq 1$

We present probability expressions for the regions shown in Fig. 3. Due to symmetry, we focus on the range where $\gamma_k > \gamma_l$. We derive the probabilities P_{A} to P_{E} as follows.

$$P_{\mathsf{A}} = \frac{e^{-\frac{1}{\bar{\gamma}}((1+\eta_0)^2-1)}}{1+\eta_0}, \quad P_{\mathsf{B}} = \frac{e^{-\frac{\eta_0}{\bar{\gamma}}}}{1+\eta_0} - \frac{e^{-\frac{1}{\bar{\gamma}}((1+\eta_0)^2-1)}}{1+\eta_0},$$

$$P_{\mathsf{C}} = \frac{1}{2}e^{-\frac{2\eta_0}{\bar{\gamma}}} - \frac{e^{-\frac{1}{\bar{\gamma}}((1+\eta_0)^2-1)}}{1+\eta_0},$$

$$P_{\mathsf{D}} = \frac{\eta_0 e^{-\frac{\eta_0}{\bar{\gamma}}}}{1+\eta_0} - e^{-\frac{2\eta_0}{\bar{\gamma}}} + \frac{e^{-\frac{\eta_0(2+\eta_0)}{\bar{\gamma}}}}{1+\eta_0}, \; P_{\mathsf{E}} = \frac{1}{2}\left(1 - e^{-\frac{\eta_0}{\bar{\gamma}}}\right)^2.$$

B. $0 < R < 1$

We present analytical probability expressions of $P_{(\mathsf{A}' \cup \mathsf{F}')}$ and $P_{\mathsf{C}'}$, whose corresponding regions are shown in Fig. 4.

To unify the two cases, as in the case of $R \geq 1$, we give the probabilities for regions $(\mathsf{A}' \cup \mathsf{F}')$ and C' $(\gamma_k > \gamma_l)$:

$$P_{(\mathsf{A}' \cup \mathsf{F}')} = \frac{1}{1+\eta_0} e^{-\frac{\eta_0^2+2\eta_0}{\bar{\gamma}}} - \frac{1-\eta_0}{2(1+\eta_0)} e^{-\frac{2\eta_0}{\bar{\gamma}(1-\eta_0)}}, \tag{49a}$$

$$P_{\mathsf{C}'} = \frac{1-\eta_0}{2(1+\eta_0)} e^{-\frac{2\eta_0}{\bar{\gamma}(1-\eta_0)}} - \left(\frac{1}{1+\eta_0} e^{-\frac{\eta_0^2+2\eta_0}{\bar{\gamma}}} - \frac{1}{2} e^{-\frac{2\eta_0}{\bar{\gamma}}}\right). \tag{49b}$$

Here, the probabilities for regions $(\breve{\mathsf{A}}' \cup \breve{\mathsf{F}}')$ and $\breve{\mathsf{C}}'$ $(\gamma_k < \gamma_l)$ are omitted due to symmetry.

APPENDIX II
DERIVATION OF $D(m, 1, L)$

A. Derivation of the Multi-Antenna Decoding Probability from the Single-Antenna Decoding Probability

For $R > 0$, we derive an exact expression for $D(m, 1, L)$, defined as the probability that a reference packet among m colliding packets is selected and decoded at the first SIC step with L receive antennas. Although multiple packets may simultaneously satisfy the decoding condition when $0 < R < 1$, only one packet is selected for decoding at the first SIC step.

We first relate $D(m, 1, L)$ to the corresponding single-antenna probability $D(m, 1, 1)$, defined as the probability that the reference packet is selected and decoded at the first SIC step at a single antenna. Since the received SNRs are independent across antennas, the probability that no packet is selected for decoding at the first SIC step at any of the L antennas is $\{1 - mD(m, 1, 1)\}^L$. Consequently, the probability that at least one packet is selected for decoding at the first SIC step at one or more of the L antennas is $1 - \{1 - mD(m, 1, 1)\}^L$.

Since the packet-selection rule is symmetric with respect to the packet labels, each of the m packets is equally likely to be selected and decoded at the first SIC step. Therefore,

$$\begin{aligned} D(m, 1, L) &= \Pr(\text{reference packet is decoded first}) \\ &= \frac{1}{m} \Pr(\text{at least one packet is decoded first}) \\ &= \frac{1}{m}\left[1 - \{1 - mD(m, 1, 1)\}^L\right]. \end{aligned} \tag{50}$$

B. Derivation of the Single-Antenna Decoding Probability

It remains to derive the single-antenna probability $D(m, 1, 1)$. Let $\mathcal{D}_i \triangleq \left\{\frac{\gamma_i}{1+\sum_{j\neq i}\gamma_j} > \eta_0\right\}$ denote the event that packet i satisfies the decoding condition. Whenever $\bigcup_{i=1}^m \mathcal{D}_i$ occurs, one of the packets satisfying the decoding condition is selected and decoded at the first SIC step.

Under the packet-label-symmetric selection rule employed by SC-SIC, the events that the individual packets are selected and decoded first are mutually exclusive and equiprobable. Therefore,

$$D(m, 1, 1) = \frac{1}{m} \Pr\left(\bigcup_{i=1}^m \mathcal{D}_i\right). \tag{51}$$

To derive $D(m, 1, 1)$, we next evaluate $\Pr\left(\bigcup_{i=1}^m \mathcal{D}_i\right)$ using the inclusion–exclusion principle. Specifically, by the exchangeability of the packet SNRs,

$$\Pr\left(\bigcup_{i=1}^m \mathcal{D}_i\right) = \sum_{s=1}^m (-1)^{s+1} \binom{m}{s} \Pr\left(\bigcap_{i=1}^s \mathcal{D}_i\right). \tag{52}$$

Here, $\Pr\left(\bigcap_{i=1}^s \mathcal{D}_i\right)$, for $1 \leq s \leq m$, denotes the probability that all packets in an arbitrary subset of s packets simultaneously satisfy the decoding condition. The remainder of this appendix derives this intersection probability.

To evaluate the intersection probability for an arbitrary subset of s packets, let $X \triangleq \sum_{i=1}^s \gamma_i$ and $Y \triangleq \sum_{i=s+1}^m \gamma_i$. Since the γ_i are independent exponentially distributed RVs with mean $\bar{\gamma}$, X follows a Gamma distribution with shape parameter s. For $1 \leq s < m$, Y follows a Gamma distribution with shape parameter $m - s$, whereas for $s = m$, $Y = 0$ deterministically [29]. The corresponding PDFs are

$$f_X(x) = \frac{x^{s-1}}{(s-1)!\,\bar{\gamma}^s} e^{-\frac{x}{\bar{\gamma}}}, \; x \geq 0, \; 1 \leq s \leq m, \tag{53}$$

$$f_Y(y) = \frac{y^{m-s-1}}{(m-s-1)!\,\bar{\gamma}^{m-s}} e^{-\frac{y}{\bar{\gamma}}}, \; y \geq 0, \; 1 \leq s < m. \tag{54}$$

Because X and Y are sums of disjoint sets of independent RVs, they are mutually independent, yielding $f_{X,Y}(x, y) = f_X(x) f_Y(y)$. By the law of total probability,

$$\Pr\left(\bigcap_{i=1}^s \mathcal{D}_i\right) = \int_0^\infty f_Y(y) \int_0^\infty \Pr\left(\bigcap_{i=1}^s \mathcal{D}_i \middle| X = x, Y = y\right)$$

$$\times f_X(x)\,dx\,dy. \tag{55}$$

For $1 \le s < m$, since Y is independent of $\gamma_1,\ldots,\gamma_s$, conditioning additionally on $Y = y$ does not affect the conditional joint PDF of $\gamma_1,\ldots,\gamma_{s-1}$ given $X = x$. Applying Bayes' rule and dividing the joint PDF of $\gamma_1,\ldots,\gamma_{s-1},X$ by the marginal PDF $f_X(x)$ yields the conditional joint PDF

$$f_{\gamma_1,\ldots,\gamma_{s-1}|X}(\gamma_1,\ldots,\gamma_{s-1}|x) = \frac{\frac{1}{\bar\gamma^s}e^{-x/\bar\gamma}}{\frac{x^{s-1}}{(s-1)!\,\bar\gamma^s}e^{-x/\bar\gamma}} = \frac{(s-1)!}{x^{s-1}}, \tag{56}$$

for $\gamma_i \ge 0$ and $\sum_{i=1}^{s-1}\gamma_i \le x$.

Conditioned on $X = x$ and $Y = y$, the decoding condition for packet $i \in \{1,\ldots,s\}$ requires $\gamma_i/(1+x+y-\gamma_i) > \eta_0$, or equivalently,

$$\gamma_i > a(x,y), \quad \text{where } a(x,y) \triangleq \frac{\eta_0(1+x+y)}{1+\eta_0}. \tag{57}$$

Substituting $\gamma_s = x - \sum_{i=1}^{s-1}\gamma_i$ into the decoding condition for packet s and rearranging yields the upper bound constraint:

$$\sum_{i=1}^{s-1}\gamma_i < x - a(x,y). \tag{58}$$

Thus, the conditional probability is obtained by integrating the conditional joint PDF over the feasible region, i.e.,

$$\Pr\left(\bigcap_{i=1}^{s}\mathcal{D}_i \middle| X=x, Y=y\right) = \int_{\mathcal{R}_s(x,y)} \frac{(s-1)!}{x^{s-1}}\,d\gamma_1\cdots d\gamma_{s-1}, \tag{59}$$

where the integration region $\mathcal{R}_s(x,y)$ is defined by the lower and upper bounds as

$$\mathcal{R}_s(x,y) \triangleq \left\{(\gamma_1,\ldots,\gamma_{s-1}) : \gamma_i > a(x,y),\ i=1,\ldots,s-1,\ \sum_{i=1}^{s-1}\gamma_i < x - a(x,y)\right\}. \tag{60}$$

By shifting each integration variable by the common lower bound, i.e., introducing $u_i = \gamma_i - a(x,y)$, the integration region $\mathcal{R}_s(x,y)$ becomes $u_i > 0$, $\sum_{i=1}^{s-1}u_i < x - sa(x,y)$, which is a standard $(s-1)$-dimensional simplex. Evaluating the integral over this region yields

$$\Pr\left(\bigcap_{i=1}^{s}\mathcal{D}_i \middle| X=x, Y=y\right) = \begin{cases} \left[\dfrac{\{1-(s-1)\eta_0\}x - s\eta_0(1+y)}{(1+\eta_0)x}\right]^{s-1}, & x > x^\star, \\ 0, & \text{otherwise}, \end{cases} \tag{61}$$

where the threshold $x^\star \triangleq \frac{s\eta_0(1+y)}{1-(s-1)\eta_0}$ is obtained from the feasibility condition $x - sa(x,y) > 0$, provided that $1-(s-1)\eta_0 > 0$. If this feasibility condition is violated, the feasible region is empty, and hence the selected s packets cannot simultaneously satisfy the decoding condition.

Substituting (61) into (55) gives

$$\Pr\left(\bigcap_{i=1}^{s}\mathcal{D}_i\right) = \left(\frac{1-(s-1)\eta_0}{1+\eta_0}\right)^{m-1} e^{-\frac{s\eta_0}{\bar\gamma\{1-(s-1)\eta_0\}}}, \tag{62}$$

for $1-(s-1)\eta_0 > 0$, and zero otherwise.

Finally, substituting (62) into (50) yields the explicit expression for $D(m,1,L)$:

$$D(m,1,L) = \frac{1}{m}\left[1 - \left\{1 - \sum_{\substack{1\le s\le m \\ 1-(s-1)\eta_0>0}} (-1)^{s+1}\binom{m}{s}\left[\frac{1-(s-1)\eta_0}{1+\eta_0}\right]^{m-1} \times e^{-\frac{s\eta_0}{\bar\gamma\{1-(s-1)\eta_0\}}}\right\}^L\right]. \tag{63}$$

For $m = 1$ and $m = 2$, (63) reduces to (12) and (13), respectively.

Appendix III
Derivation of the High-SNR Error Floor

A. Exact High-SNR Error Floor for $m = 2$

We first derive the exact high-SNR error floor for the two-packet case in (26). For $\eta_0 > 1$, the high-SNR limits of the region probabilities are $\lim_{\bar\gamma\to\infty} P_{\mathsf{A}} = \frac{1}{1+\eta_0}$, $\lim_{\bar\gamma\to\infty} P_{\mathsf{C}} = \frac{\eta_0-1}{2(1+\eta_0)}$, and all the other region probabilities tend to zero as $\bar\gamma \to \infty$. Taking the high-SNR limit of $D(2,1,L)$ in (13) and $D(2,2,L)$ in (15) yields

$$\lim_{\bar\gamma\to\infty} D(2,1,L) = \lim_{\bar\gamma\to\infty} D(2,2,L) = \frac{1}{2}\left\{1 - \left(\frac{\eta_0-1}{\eta_0+1}\right)^L\right\}.$$

Using (10), we obtain

$$\lim_{\bar\gamma\to\infty} \bar\epsilon_{2,L} = \left(\frac{\eta_0-1}{\eta_0+1}\right)^L, \qquad \eta_0 > 1.$$

B. Approximate High-SNR Error Floor for $m \ge 3$

We next derive the approximate high-SNR error floor for $m \ge 3$ over the entire range $R > 0$, or equivalently, $\eta_0 > 0$. For each integer s satisfying $1-(s-1)\eta_0 > 0$, we have $\lim_{\bar\gamma\to\infty} e^{-\frac{s\eta_0}{\bar\gamma\{1-(s-1)\eta_0\}}} = 1$.

Therefore, taking the high-SNR limit of (63) yields

$$\lim_{\bar\gamma\to\infty} D(r,1,L) = \frac{1}{r}\left\{1 - \left[1 - \sum_{\substack{1\le s\le r \\ 1-(s-1)\eta_0>0}} (-1)^{s+1}\binom{r}{s}\left(\frac{1-(s-1)\eta_0}{1+\eta_0}\right)^{r-1}\right]^L\right\} = \frac{1}{r}\Phi_{r,L}(\eta_0), \tag{64}$$

where $\Phi_{r,L}(\eta_0)$ is defined in (28). Moreover, the exact two-packet result derived in the preceding subsection gives $\lim_{\bar\gamma\to\infty} D(2,L) = \Phi_{2,L}(\eta_0)$.

Taking the high-SNR limit of the closed-form approximation in (20) and substituting (64), we obtain

$$\lim_{\bar{\gamma}\to\infty} \widetilde{D}(m,L) = \frac{1}{m}\left[\sum_{j=3}^{m}\prod_{r=j}^{m}\Phi_{r,L}(\eta_0) + 2\prod_{r=2}^{m}\Phi_{r,L}(\eta_0)\right], \tag{65}$$

where we used $\frac{(m-1)!}{(j-1)!}\prod_{r=j}^{m} r^{-1} = 1/m$ and $(m-1)!\prod_{r=3}^{m} r^{-1} = 2/m$.

Finally, since

$$\widetilde{\epsilon}_{m,L}^{\text{floor}}(\eta_0) = 1 - \lim_{\bar{\gamma}\to\infty} \widetilde{D}(m,L), \tag{66}$$

(27) follows immediately.

REFERENCES

[1] W. Jiang, B. Han, M. A. Habibi, and H. D. Schotten, "The road towards 6G: A comprehensive survey," *IEEE Open J. Commun. Soc.*, vol. 2, pp. 334–366, 2021.

[2] G. Liva and Y. Polyanskiy, "Unsourced multiple access: A coding paradigm for massive random access," *Proc. IEEE*, vol. 112, no. 9, pp. 1214–1229, 2024.

[3] A. Munari, "Modern random access: An age of information perspective on irregular repetition slotted ALOHA," *IEEE Trans. Commun.*, vol. 69, no. 6, pp. 3572–3585, 2021.

[4] N. Abramson, "The ALOHA system: Another alternative for computer communications," in *Proc. AFIPS Fall Joint*, 1970, pp. 281–285.

[5] E. Casini, R. De Gaudenzi, and O. Del Rio Herrero, "Contention resolution diversity slotted ALOHA (CRDSA): An enhanced random access scheme for satellite access packet networks," *IEEE Trans. Wireless Commun.*, vol. 6, no. 4, pp. 1408–1419, 2007.

[6] F. Clazzer, E. Paolini, I. Mambelli, and Č. Stefanović, "Irregular repetition slotted ALOHA over the Rayleigh block fading channel with capture," in *Proc. IEEE Int. Conf. Commun. (ICC)*, 2017, pp. 1–6.

[7] Y. Takahashi, G. Song, T. Kimura, and J. Cheng, "Irregular repetition slotted ALOHA over Rayleigh block fading channels: Bounds and threshold saturation via spatial coupling," *IEEE Access*, vol. 11, pp. 106 528–106 543, 2023.

[8] G. Liva, E. Paolini, M. Lentmaier, and M. Chiani, "Spatially-coupled random access on graphs," in *Proc. IEEE Int. Symp. Inf. Theory (ISIT)*, 2012, pp. 478–482.

[9] G. Liva, "Graph-based analysis and optimization of contention resolution diversity slotted ALOHA," *IEEE Trans. Commun.*, vol. 59, no. 2, pp. 477–487, 2011.

[10] E. Paolini, Č. Stefanović, G. Liva, and P. Popovski, "Coded random access: Applying codes on graphs to design random access protocols," *IEEE Commun. Mag.*, vol. 53, no. 6, pp. 144–150, 2015.

[11] G. Liva, E. Paolini, Č. Stefanović, and A. G. i Amat, "Coded slotted ALOHA over the on-off fading channel: Performance bounds," in *Proc. 53rd Asilomar Conf. Signals, Syst., Comput.*, 2019, pp. 31–35.

[12] Y. Takahashi, I. Masuo, G. Song, T. Kimura, and J. Cheng, "Coded slotted ALOHA over Rayleigh block fading channels: BP threshold and converse bound," in *Proc. 2023 IEEE Globecom Wkshps. (GC Wkshps.)*, 2023, pp. 1523–1528.

[13] V. Tralli and E. Paolini, "IRSA-based random access over the Gaussian channel," *IEEE Trans. Inf. Theory*, vol. 70, no. 6, pp. 4117–4139, 2024.

[14] D. Jakovetić, D. Bajović, D. Vukobratović, and V. Crnojević, "Cooperative slotted ALOHA for multi-base station systems," *IEEE Trans. Commun.*, vol. 63, no. 4, pp. 1443–1456, 2015.

[15] C.-H. Yu, L. Huang, C.-S. Chang, and D.-S. Lee, "Poisson receivers: A probabilistic framework for analyzing coded random access," *IEEE/ACM Trans. Netw.*, vol. 29, no. 2, pp. 862–875, 2021.

[16] E. Recayte and C. Amatetti, "Multi-satellite NOMA-irregular repetition slotted ALOHA for IoT networks," in *Proc. IEEE Int. Conf. Commun. (ICC)*, 2025, pp. 5204–5209.

[17] S. Ogata and K. Ishibashi, "Frameless ALOHA with multiple base stations," in *Proc. 49th Asilomar Conf. Signals, Syst., Comput.*, 2015, pp. 47–51.

[18] S. Ogata, K. Ishibashi, and G. Abreu, "Multi-access diversity gain via multiple base station cooperation in frameless ALOHA," in *Proc. IEEE 18th Int. Workshop Signal Process. Adv. Wireless Commun. (SPAWC)*, 2017, pp. 1–5.

[19] S. Ogata, K. Ishibashi, and G. T. de Abreu, "Optimized frameless ALOHA for cooperative base stations with overlapped coverage areas," *IEEE Trans. Wireless Commun.*, vol. 17, no. 11, pp. 7486–7499, 2018.

[20] X. Li, L. Zhao, and T. Yang, "Pilot-free slotted ALOHA for massive access," *IEEE Trans. Veh. Technol.*, vol. 73, no. 9, pp. 13 903–13 908, 2024.

[21] T.-H. Liu, C.-H. Yu, Y.-J. Lin, C.-M. Chang, C.-S. Chang, and D.-S. Lee, "ALOHA receivers: A network calculus approach for analyzing coded multiple access with SIC," *IEEE/ACM Trans. Netw.*, vol. 30, no. 2, pp. 840–854, 2022.

[22] S. Qin, G. Ren, and Y. He, "NOMAC-IRSA: A novel grant-free access scheme for distributed random access systems," *IEEE Trans. Veh. Technol.*, pp. 1–16, 2026.

[23] C. R. Srivatsa and C. R. Murthy, "On the impact of channel estimation on the design and analysis of IRSA based systems," *IEEE Trans. Signal Process.*, vol. 70, pp. 4186–4200, 2022.

[24] T. Richardson and R. Urbanke, *Modern Coding Theory*. Cambridge: Cambridge University Press, 2008.

[25] E. Paolini, G. Liva, and M. Chiani, "Graph-based random access for the collision channel without feedback: Capacity bound," in *Proc. 2011 IEEE Global Telecommun. Conf. (GLOBECOM)*, 2011, pp. 1–5.

[26] A. Ashikhmin, G. Kramer, and S. ten Brink, "Extrinsic information transfer functions: Model and erasure channel properties," *IEEE Trans. Inf. Theory*, vol. 50, no. 11, pp. 2657–2673, 2004.

[27] A. Mengali, R. De Gaudenzi, and Č. Stefanović, "On the modeling and performance assessment of random access with SIC," *IEEE J. Sel. Areas Commun.*, vol. 36, no. 2, pp. 292–303, 2018.

[28] Z. Zhang, Y. Li, G. Song, C. Yuen, and Y. L. Guan, "Random NOMA with cross-slot successive interference cancelation packet recovery," *IEEE Wireless Commun. Lett.*, vol. 9, no. 7, pp. 1065–1069, 2020.

[29] S. M. Ross, *A First Course in Probability*, 6th ed. Upper Saddle River, NJ, USA: Prentice Hall, 1998.

[30] N. Zhang, J. Wang, G. Kang, and Y. Liu, "Uplink nonorthogonal multiple access in 5G systems," *IEEE Commun. Lett.*, vol. 20, no. 3, pp. 458–461, 2016.

[31] D. Tse and P. Viswanath, *Fundamentals of Wireless Communication*. USA: Cambridge University Press, 2005.

[32] R. Storn and K. Price, "Differential evolution–a simple and efficient heuristic for global optimization over continuous spaces," *J. Glob. Optim.*, vol. 11, no. 4, pp. 341–359, 1997.

[33] Č. Stefanović, E. Paolini, and G. Liva, "Asymptotic performance of coded slotted ALOHA with multipacket reception," *IEEE Commun. Lett.*, vol. 22, no. 1, pp. 105–108, 2018.

[34] S. Kudekar, T. J. Richardson, and R. L. Urbanke, "Threshold saturation via spatial coupling: Why convolutional LDPC ensembles perform so well over the BEC," *IEEE Trans. Inf. Theory*, vol. 57, no. 2, pp. 803–834, 2011.